PRACTICE SHEET

1. The multiplicative inverse of $\sqrt{5}+3i$ is

(a) $\frac{\sqrt{5}}{14}+\frac{3i}{14}$ (b) $\frac{\sqrt{5}}{14}-\frac{3i}{14}$

(c) $\frac{\sqrt{5}}{13}-\frac{3i}{13}$ (d) $\frac{\sqrt{5}}{13}+\frac{3i}{13}$

2. If $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3 = x+iy$, then (x, y) is equal to

(a) (0, –2) (b) (–2, 0)
(c) (0, 2) (d) (2, 0)

3. If $\frac{(1+i)^2}{2-i}=x+iy$, then the value of x + y is equal to

(a) 5/2 (b) –2/5
(c) 2/5 (d) –5/2

4. The value of $1+i^2+i^4+i^6+\ldots\ldots i^{2n}$ is
(a) Positive
(b) Negative
(c) Zero
(d) Cannot be determined

5. The value of sum $\sum_{n=1}^{13}\left(i^n+i^{n+1}\right)$ where $i=\sqrt{-1}$, is equal to

(a) i (b) i – 1
(c) –i (d) 0

6. If $\left(\frac{1+i}{1-i}\right)^x=1$, then
(a) x = 2n + 1
(b) x = 4n
(c) x = 2n
(d) x = 4n + 1, where $n \in N$

7. The real value of θ for which the expression $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is a real number is

(a) $n\pi+\frac{\pi}{4}$ (b) $n\pi+(-1)^n\frac{\pi}{4}$

(c) $2n\pi\pm\frac{\pi}{4}$ (d) None of these

8. Given $z=\frac{q+ir}{1+p}$, then $\frac{p+iq}{1+r}=\frac{1+iz}{1-iz}$, if

(a) $p^2+q^2+r^2=1$ (b) $p^2+q^2+r^2=2$
(c) $p^2+q^2-r^2=1$ (d) None of these

9. If z = x + iy lies in the third quadrant, then $\frac{\bar{z}}{z}$ also lies in the third quadrant, if
(a) x > y > 0 (b) x < y < 0
(c) y < x < 0 (d) y > x > 0

10. The conjugate of $\frac{2-i}{(1-2i)^2}$ is

(a) $\frac{2}{25}-\frac{i11}{25}$ (b) $\frac{-2}{25}-\frac{i11}{25}$

(c) $\frac{-2}{25}+\frac{i11}{25}$ (d) $\frac{2}{25}+\frac{i11}{25}$

11. If $\frac{z-1}{z+1}$ is a purely imaginary number ($z\neq-1$), then the value of |z| is
(a) –1 (b) 1
(c) 2 (d) –2

12. If α and β are different complex numbers with |β| = 1, then $\left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right|$ is equal to
(a) 0 (b) 1/2
(c) 1 (d) 2

13. If |z + 1| = z + 2(1 + i), then the value of z is

(a) $\frac{1}{2}-2i$ (b) $\frac{1}{2}+2i$

(c) $\frac{1}{2}-3i$ (d) $\frac{1}{3}-2i$

14. If z is a complex number, then $(\bar{z}^{-1})(\bar{z})$ is equal to
(a) 1 (b) –1
(c) 0 (d) None of these

15. The complex number z which satisfies the condition $\left|\frac{i+z}{i-z}\right|=1$ lies on
(a) Circle $x^2+y^2=1$
(b) The axis
(c) The y-axis
(d) The line x + y = 1

16. $|z_1+z_2=|z_1|+|z_2|$ is possible, if

(a) $z_2=\bar{z}_1$ (b) $z_2=\frac{1}{z_1}$

(c) arg (z_1) =arg (z_2) (d) $|z_1|=|z_2|$

17. If z is a complex number, then |3z – 1| = 3|z – 2| represents
(a) Y-axis
(b) A circle
(c) X-axis
(d) A line parallel to y-axis

18. arg $(\bar{z})$ is equal to
(a) π – arg(z) (b) 2π – arg(z)
(c) π + arg(z) (d) 2π + arg(z)

19. If arg (z – 1) = arg (z + 3i), then x – 1 : y is equal to
(a) 3 : 1 (b) 1 : 3
(c) 3 : 2 (d) 2 : 3

20. If $f(z)=\frac{7-z}{1-z^2}$, where z = 1 + 2i, then |f(z)| is

(a) $\frac{|z|}{2}$ (b) |z|

(c) 2 |z| (d) None of these

21. If $\tan^{-1}(\alpha+i\beta)=x+iy$, then x is equal to

(a) $\frac{1}{2}\tan^{-1}\left(\frac{2\alpha}{1-\alpha^2-\beta^2}\right)$

(b) $\frac{1}{2}\tan^{-1}\left(\frac{2\alpha}{1+\alpha^2+\beta^2}\right)$

(c) $\tan^{-1}\left(\frac{2\alpha}{1-\alpha^2-\beta^2}\right)$

(d) None of the above

22. If $\bar{z}$ be the conjugate of the complex number z, then which of the following relations is false?

(a) $|z|=|\bar{z}|$ (b) $z.\bar{z}=|\bar{z}|^2$

(c) $\overline{z_1+z_2}=\bar{z}_1+\bar{z}_2$ (d) arg(z) = arg($\bar{z}$)

23. If z_1, z_2 and z_3, z_4 are two pairs of conjugate complex numbers, then $\arg\left(\frac{z_1}{z_4}\right)+\arg\left(\frac{z_2}{z_3}\right)$ is equal to

(a) 0 (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

24. The argument of $\frac{(1-i\sqrt{3}}{(1+i\sqrt{3})}$ is

(a) 60° (b) 120°
(c) 210° (d) 240°

25. If z and w are two non-zero complex numbers such that |zw| = 1 and arg (z) – arg (w) = $\frac{\pi}{2}$, $\bar{z}w$ is equal to

(a) 1 (b) –i
(c) i (d) –1

26. If $|z + 4| \le 3$, then the greatest and the least value of $|z + 1|$ are

(a) 6, –6 (b) 6, 0
(c) 7, 2 (d) 0, –1

27. If z is a complex number, then the minimum value of |z| + |z – 1| is

(a) 1 (b) 0
(c) $\frac{1}{2}$ (d) None of these

28. If z_1, z_2 and z_3 be three complex numbers such that $|z_1 + 1| \le 1$, $|z_2 + 2| \le 2$ and $|z_3 + 4| \le 4$, then the maximum value of $|z_1| + |z_2| + |z_3|$ is

(a) 7 (b) 10
(c) 12 (d) 14

29. The maximum value of |z| where z satisfies the condition $\left|z+\frac{2}{z}\right|=2$ is

(a) $\sqrt{3}-1$ (b) $\sqrt{3}+1$

(c) $\sqrt{3}$ (d) $\sqrt{2}+\sqrt{3}$

30. What is the conjugate of $\left(\frac{1+2i}{2+i}\right)^2$?

(a) $\frac{7}{25}+i\frac{24}{25}$ (b) $-\frac{7}{25}-i\frac{24}{25}$

(c) $-\frac{7}{25}+i\frac{24}{25}$ (d) $\frac{7}{25}-i\frac{24}{25}$

31. What is the value of $1 + i^2 + i^4 + i^6 + \ldots\ldots i^{100}$, where $i=\sqrt{-1}$

(a) 0 (b) 1
(c) –1 (d) None of these

32. If ω is a complex cube root of unity, then what is $\omega^{10} + \omega^{-10}$ equal to?

(a) 2 (b) –1
(c) –2 (d) 1

33. What is $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^6$ equal to

(a) –1 (b) 0
(c) 1 (d) 2

34. What is the modulus of $\left|\frac{1+2i}{1-(1-i)^2}\right|$?

(a)1 (b)√5
(c)√3 (d) 5

35. What is the value of $\left(\frac{i+\sqrt{3}}{-i+\sqrt{3}}\right)^{200}+\left(\frac{i-\sqrt{3}}{i+\sqrt{3}}\right)^{200}+1$?

(a) –1 (b) 0
(c) 1 (d) 2’

36. If $x^2 + y^2 = 1$, then what is $\frac{1+x+iy}{1+x-iy}$ equal to?

(a) x–iy (b) x+iy
(c) 2x (d) –2iy

37. What is the modulus of $\frac{1+2i}{1-(1+i)^2}$ equal to?

(a) 5 (b) 4
(c) 3 (d) +1

38. What is the value of $(-\sqrt{-1})^{4n+3} + (i^{41} + i^{-257})$, where $n \in N$?

(a) 0 (b) 1
(c) i (d) –i

39. If α is a complex number such that $\alpha^2 + \alpha + 1 = 0$, then what is α^{31} equal to?

(a) α (b) α^2
(c) 0 (d) 1

40. If ω is the cube root of unity, then what is the conjugate of $2\omega^2 + 3i$?

(a) 2ω–3i (b) 3ω+2i
(c) 2ω + 3i (d) 3ω - 2i

41. What is $\left(\sqrt{3}+i\right)/\left(1+\sqrt{3}i\right)$ equal to?

(a) 1+i (b) 1–i
(c)√3(1–i)/2 (d) (√3-i)/2

42. If 2x = 3 + 5i, then what is the $2x^3 + 2x^2 - 7x + 72$?

(a) 4 (b) –4
(c) 8 (d) –8

43. Match List I with List II and select the correct answer using the code given below the lists:

List-I	List-II
A. A cube root of unity	(i) $-2(1+i)$
B. A square root of –1	(ii) 2i
C. Cube of 1-i	(iii) i
D. Square of 1+i	(iv) $-\frac{1}{2}\left(1+i\sqrt{3}\right)$

Codes:
(a) A-iv, B-i, C-iii, D-ii
(b) A-ii, B-i, C-iii, D-iv
(c) A-iv, B-iii, C-i, D-ii
(d) A-ii, B-iii, C-i, D-iv

44. For positive whole number of n, what is the value of i^{4n+1}?

(a) 1 (b) –1
(c) i (d) –i

45. If ω is complex cube root, then what is the value of $1-\frac{1}{(1+\omega)}-\frac{1}{\left(1+\omega^2\right)}$?

(a) 1 (b) 0
(c) ω (d) ω2

46. A straight line is passing through the points represented by the complex numbers a + ib and $\frac{1}{-a+ib}$, where $(a, b)\neq(0,0)$. Which one of the following is correct?
(a) It passes through the origin
(b) It is parallel to the a-axis
(c) It is parallel to the y-axis
(d) It passes through (0, b)

47. Which one of the following is correct? If z and w are complex numbers and $\bar{w}$ denotes the conjugate of w, then $|z+w|=|z-w|$ holds only, if:
(a) z = 0 or w = 0
(b) z = 0 and w = 0
(c) $z.\bar{w}$ is purely real
(d) $z.\bar{w}$ is purely imaginary

48. What is the square root of $\frac{1}{2}-i\frac{\sqrt{3}}{2}$?

(a) $\pm\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)$ (b) $\pm\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)$
(c) $\pm\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)$ (d) $\pm\left(\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)$

49. Let C be the set of complex number and z_1, z_2 are in C.
I. $\arg(z_1) = \arg(z_2) \Rightarrow z_1 = z_2$
II. $|z_1| = |z_2| \Rightarrow z_1 = z_2$
Which of the statements given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

50. If 1, ω and ω^2 are the three cube roots of unity, then what is the value of $\frac{\left(aw^6+b\omega^4+c\omega^2\right)}{\left(b+c\omega^{10}+a\omega^8\right)}$?

(a) a/b (b) b
(c) ω (d) ω^2

ANSWER KEY																			
1.	b	**2.**	a	**3.**	c	**4.**	d	**5.**	b	**6.**	b	**7.**	d	**8.**	a	**9.**	c	**10.**	b
11.	b	**12.**	c	**13.**	a	**14.**	a	**15.**	b	**16.**	c	**17.**	d	**18.**	b	**19.**	b	**20.**	a
21.	a	**22.**	d	**23.**	a	**24.**	d	**25.**	b	**26.**	b	**27.**	a	**28.**	b	**29.**	b	**30.**	d
31.	b	**32.**	b	**33.**	c	**34.**	a	**35.**	b	**36.**	b	**37.**	d	**38.**	c	**39.**	a	**40.**	a
41.	d	**42.**	a	**43.**	c	**44.**	c	**45.**	b	**46.**	a	**47.**	d	**48.**	b	**49.**	d	**50.**	c

Solutions

Sol.1. (b)
Let

$z=\sqrt{5}+3i$, then its multiplicative inverse is

$$\frac{1}{z}=\frac{1}{\sqrt{5}+3i}=\frac{1}{\sqrt{5}+3i}\times\frac{\sqrt{5}-3i}{\sqrt{5}-3i}$$

$$\frac{\sqrt{5}-3i}{5-9i^2}\quad[\because (a+b)(a-b)=a^2-b^2]$$

$$=\frac{\sqrt{5}-3i}{5+9}=\frac{\sqrt{5}-3i}{14}\quad(\because i^2=-1)$$

$$=\frac{\sqrt{5}}{14}-\frac{3i}{14}$$

Sol.2. (a)

Given, $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3=x+iy$

$$=\frac{(1+i)^2}{1^2+(i)^2}=\frac{1^2+(i)^2+2i}{1-(-1)}$$

$$=\frac{1-1+2i}{1+1}=\frac{2i}{2}=i$$

∴

$$\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3=(i)^3-\left(\frac{1}{i}\right)^3=-i-\frac{1}{(-i)}$$

$$=-i+\frac{1}{i}\times\frac{i}{i}=-1+\frac{i}{-1}=-2i$$

∴ –2i = x + iy [∵ From eq. (i)]
On comparing the real and imaginary part both side, we get
x = 0, y = – 2

Sol.3. (c)

Given, $\frac{(1+i)^2}{2-i}=x+iy$

$$\therefore\ \frac{1^2+(i)^2+2i}{2-i}\times\frac{2+i}{2+i}=x+iy$$

$$\Rightarrow \frac{(1-1+2i)(2+i)}{2^2-(i)^2}=x+iy$$

$$\Rightarrow \frac{4i-2}{4+1}=x+iy$$

$$\Rightarrow \frac{4i}{5}-\frac{2}{5}=x+iy$$

On comparing real and imaginary parts, we get

$$x=-\frac{2}{5},\ y=\frac{4}{5}$$

$$\therefore\ x+y=\frac{2}{5}+\frac{4}{5}=\frac{2}{5}$$

Sol.4. (d)
Let $S = 1 + i^2 + i^4 + i^6 + \ldots + i^{2n}$
$= 1 - 1 + 1 - 1 + 1 - \ldots\ldots + (-1)^n$
The value of S depends on n.
∴ The value cannot be determined

Sol.5. (b)

As sum of any four consecutive powers of iota is zero

$\therefore \sum_{n=1}^{13}(i^n + i^{n+1}) = (i + i^2 + \ldots + i^{13}) + (i^2 + i^3 + \ldots + i^{14})$

$= i + i^2 = i - 1$

Sol.6. (b)

$$\left(\frac{1+i}{1-i}\right)^x = \left[\frac{(1+i)}{(1-i)}\times\frac{1+i}{1+i}\right]^x$$

$$= \left[\frac{(1+i)^2}{1^2-i^2}\right]^x = \left[\frac{1^2+i^2+2i}{1+1}\right]^x$$

$$= \left(\frac{1-1+2i}{2}\right)^x = i^x$$

But it is given $i^x = 1$

$\Rightarrow x = 4n$, where $n \in N$

Sol.7. (d)

$$\frac{1+i\cos\theta}{1-2i\cos\theta} = \frac{1+i\cos\theta}{1-2i\cos\theta}\times\frac{1+2i\cos\theta}{1+2i\cos\theta}$$

$$= \frac{1+3i\cos\theta+2i^2\cos^2\theta}{1^2-(2i\cos\theta)^2}$$

$$= \frac{1-2\cos^2\theta}{1+4\cos^2\theta} + \frac{i3\cos\theta}{1+4\cos^2\theta}$$

$\Rightarrow \cos\theta = 0$

$\Rightarrow \theta = 2n\pi \pm \pi/2$

Sol.8. (a)

We have, $z = \frac{q+ir}{1+p}$ $\quad\therefore iz = \frac{-r+iq}{1+p}$

By componendo and dividendo

$$\frac{1+iz}{1-iz} = \frac{1+p-r+iq}{1+p+r-iq}$$

$\therefore \frac{p+iq}{1+r} = \frac{1+iz}{1-iz}$, if $\frac{p+iq}{1+r} = \frac{1+p-r+iq}{1+p+r-iq}$

$\Rightarrow p(1 + p + r) + q^2 + i\{q(1 + p + r) - pq\}$

$= (1 + r)(1+ p - r) + iq(1 + r)$

$\Rightarrow p(1 + p + r) + q^2 = (1 + r)(1 + p - r)$

and $q(1 + p + r) - pq = q(1 + r)$ (this is obviously true)

$\therefore$ The condition is

$p(1 + p + r) + q^2 = (1 + r)(1 + p - r)$

$\Rightarrow p + p^2 + pr + q^2 = 1 + p - r + r + pr - r^2$

$\Rightarrow p^2 + q^2 + r^2 = 1$

Sol.9. (c)

$$\frac{\bar{z}}{z} = \frac{\overline{(x+iy)}}{x+iy} = \frac{x-iy}{x+iy}\times\frac{x-iy}{x-iy}$$

$$= \frac{x^2-y^2-2ixy}{x^2-i^2y^2} = \frac{x^2-y^2-2ixy}{x^2+y^2}$$

$$= \frac{x^2-y^2}{x^2+y^2} + \frac{i(-2xy)}{x^2+y^2}$$

Since, $\frac{\bar{z}}{z}$ lies in third quadrant, therefore its real and imaginary parts should be negative

Sol.10. (b)

Let $z = \frac{2-i}{(1-2i)^2}$

$\therefore$ Conjugate of $z = \bar{z} = \frac{2+i}{(1+2i)^2}$ ($\because$ replacing i by –i)

$$\therefore \bar{z} = \frac{2+i}{1^2-4+4i} = \frac{2+i}{4i-3}\times\frac{4i+3}{4i+3}$$

$$= \frac{11i+2}{(4i)^2-(3)^2} = \frac{11i+2}{-16-9} = \frac{-2}{25} - i\frac{11}{25}$$

Sol.11. (b)

Let $z = x + iy$

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1} = \frac{(x-1)+iy}{(x+1)+iy}\times\frac{(x+1)-iy}{(x+1)-iy}$$

$$= \frac{(x-1)(x+1)-iy(x-1)+iy(x+1)-i^2y^2}{(x+1)^2-i^2y^2}$$

$$= \frac{x^2-1+iy(x+1-x+1)+y^2}{(x+1)^2+y^2}$$

$$\Rightarrow \frac{z-1}{z+1} = \frac{(x^2+y^2-1)}{(x+1)^2+y^2} + \frac{i(2y)}{(x+1)^2+y^2}$$

$\because \frac{z-1}{z+1}$ is purely imaginary

$$\therefore \text{Re}\left(\frac{z-1}{z+1}\right) = 0$$

$$\Rightarrow \frac{x^2+y^2-1}{(x+1)^2+y^2} = 0$$

$\Rightarrow x^2 + y^2 - 1 = 0$

$\Rightarrow x^2 + y^2 = 1$

$\Rightarrow |z^2| = 1 \Rightarrow |z| = 1$

Sol.12. (c)

Given that, $|\beta| = 1$

$$\therefore \left|\frac{\beta-\alpha}{1-\bar{\alpha}\beta}\right| = \left|\frac{\beta-\alpha}{\beta\bar{\beta}-\bar{\alpha}\beta}\right|$$

$$= \left|\frac{\beta-\alpha}{\beta(\bar{\beta}-\bar{\alpha})}\right| = \frac{1}{|\beta|}\left|\frac{\beta-\alpha}{(\bar{\beta}-\bar{\alpha})}\right|$$

$$= \frac{1}{|\beta|} = 1 \qquad (\because |z| = |\bar{z}|)$$

Sol.13. (a)

Given, $|z + 1| = z + 2(1 + i)$ …(i)

By taking option (a),

Put $z = \frac{1}{2} - 2i$ in Eq. (i), we get

$$\text{LHS} = \left|\frac{1}{2} - 2i + 1\right| = \left|\frac{3}{2} - 2i\right|$$

$$= \sqrt{\frac{9}{4}+4} = \frac{\sqrt{25}}{2} = \frac{5}{2}$$

$$\text{RHS} = \left(\frac{1}{2} - 2i\right) + 2 + 2i = \frac{5}{2}$$

$\therefore$ LHS = RHS

Hence, option (a) is correct

Sol.14. (a)

Let $z = x + iy$

$\Rightarrow \bar{z} = x - iy$

and $(\bar{z}^{-1}) = \frac{1}{x-iy} = \frac{x+iy}{x^2+y^2}$

$$\therefore (\bar{z}^{-1})\bar{z} = \frac{x+iy}{x^2+y^2}\times(x-iy) = 1$$

Sol.15. (b)

Given, $\left|\frac{i+z}{1-z}\right| = 1$

Let $z = x + iy$

$$\therefore \left|\frac{i+x+iy}{i-(x+iy)}\right| = 1$$

$$\Rightarrow \left|\frac{x+i(1+y)}{-x+(1-y)}\right| = 1$$

$$\Rightarrow \sqrt{x^2+(1+y)^2} = \sqrt{(-x)^2+(1-y)^2}$$

$\Rightarrow x^2 + 1^2 + y^2 + 2y = x^2 + 12 + y^2 - 2y$

$\Rightarrow 4y = 0 \Rightarrow y = 0$

Hence, the given condition lies on x-axis

Sol.16. (c)

Let $z_1 = r_1 (\cos\theta_1 + i\sin\theta_2)$, $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$

$\because |z_1 + z_2| = |z_1| + |z_2|$

$|(r_1\cos\theta_1 + ir_1\sin\theta_1 + r_2\sin\theta_2)|$

$= |r_1\cos\theta_1 + ir_1\sin\theta_1| + |r_2\cos\theta_2 + ir_2\sin\theta_2|$

$$\Rightarrow \sqrt{r_1^2+r_2^2+2r_1r_2\cos(\theta_1-\theta_2)}$$

$$= \sqrt{r_1^2\cos^2\theta_1 + r_1^2\sin^2\theta_1} + \sqrt{r_2^2\cos^2\theta_2 + r_2^2\sin^2\theta_2}$$

$$\Rightarrow \sqrt{r_1^2+r_2^2+2r_1r_2\cos(\theta_1-\theta_2)} = r_1 + r_2$$

$$\Rightarrow r_1^2+r_2^2+2r_1r_2\cos(\theta_1-\theta_2) = r_1^2+r_2^2+2r_1r_2$$

$\Rightarrow \cos(\theta_1 - \theta_2) = 1$

$\Rightarrow \theta_1 - \theta_2 = 0$

$\Rightarrow \theta_1 = \theta_2$

$\Rightarrow \arg(z_1) = \arg(z_2)$

Sol.17. (d)

$|3z - 1| = 3|z - 2|$

$$\Rightarrow \left|z - \frac{1}{3}\right| = |z-2|$$

$\Rightarrow$ z is perpendicular bisector of $\left(\frac{1}{3}, 0\right)$ and (2, 0)

$$\Rightarrow x = \frac{7}{6}$$

Sol.18. (b)

Let $z = x + iy$

$$\therefore \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

Then, arg

$$(\bar{z}) = \tan^{-1}\left(-\frac{y}{x}\right) = 2\pi - \tan^{-1}\frac{y}{x}$$

$= 2\pi - \arg(z)$

Since, in argment of a conjugate of a complex, the real axis is unaltered but imginary axis be changed, hence it is given by $2\pi - \arg(z)$

Sol.19. (b)

We have, $\arg(z - 1) = \arg(z + 3i)$

On putting $z = x + iy$

$\Rightarrow \arg(x + iy - 1) = \arg(x + iy + 3i)$

$\Rightarrow \arg[(x-1)+iy] = \arg[x+i(y+3)]$

$$\Rightarrow \tan^{-1}\frac{y}{x-1} = \tan^{-1}\frac{y+3}{x}$$

$$\Rightarrow \frac{y}{x-1} = \frac{y+3}{x} \Rightarrow xy = (x-1)(y+3)$$

$\Rightarrow xy = xy + 3x - y - 3$
$\Rightarrow 0 = 3(x-1) - y$

$$\Rightarrow y = 3(x-1) \Rightarrow \frac{x-1}{y} = \frac{1}{3}$$

$\Rightarrow (x-1) : y = 1 : 3$

Sol.20. (a)

Given, $f(z) = \frac{7-z}{1-z^2}$ and $z = 1 + 2i$

$$\therefore f(z) = \frac{7-(1+2i)}{1-(1+2i)^2}$$

$$= \frac{6-2i}{1-(1-4+4i)} = \frac{6-2i}{4-4i}$$

$$= \frac{6-2i}{4(1-i)} \times \frac{1-i}{(1+i)} = \frac{6+4i+2}{4(1^2-i^2)}$$

$$= \frac{8+4i}{4(2)} = \frac{1}{2}(2+i)$$

$$|f(z)| = \frac{\sqrt{4+1}}{2} = \frac{\sqrt{5}}{2}$$

$$= \frac{|z|}{2} \quad (\because z = 1+2i, \text{ given } \Rightarrow |z| = \sqrt{5})$$

Sol.21. (a)

We have, $\tan-1(\alpha + i\beta) = x + iy$
$\Rightarrow \alpha + i\beta = \tan(x + iy)$
Taking conjugate, we get
$(\alpha - i\beta) = \tan(x - iy)$
$\therefore \tan 2x = \tan[(x + iy) + (x - iy)]$

$$\Rightarrow \tan 2x = \frac{(\alpha+i\beta)+(\alpha-i\beta)}{1-(\alpha+i\beta)(\alpha-i\beta)}$$

$$= \frac{2\alpha}{1-(\alpha^2+\beta^2)}$$

$$\therefore x = \frac{1}{2}\tan^{-1}\left(\frac{2\pi}{1-\alpha^2-\beta^2}\right)$$

Sol.22. (d)

Let $z = x + iy \quad \Rightarrow \bar{z} = x - iy$

Since, $\arg(\bar{z}) = \tan^{-1}\frac{y}{x}$

and $\arg(\bar{z}) = \tan^{-1}\left(\frac{-y}{x}\right)$

$\Rightarrow \arg(z) \neq \arg(\bar{z})$

Sol.23. (a)

We have, $z_2 = \bar{z}_1$ and $z_4 = \bar{z}_3$

Therefore, $z_1z_2 = |z_1|^2$ and $z_3z_4 = |z_3|^2$

Now, $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right) = \arg\left(\frac{z_1z_2}{z_4z_3}\right)$

$$= \arg\left(\frac{|z_1|^2}{|z_3|^2}\right) = \arg\left(\left|\frac{z_1}{z_2}\right|^2\right) = 0$$

($\because$ argument of positive real number is zero)

Sol.24. (d)

Let $z = \frac{1-i\sqrt{3}}{1+i\sqrt{3}}$

$$= \frac{1-i\sqrt{3}}{1+i\sqrt{3}} \times \frac{1-i\sqrt{3}}{1-i\sqrt{3}} = \frac{-1-i\sqrt{3}}{2}$$

$$\therefore \arg(z) = \theta = \tan^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right) = \tan^{-1}\left(\sqrt{3}\right) = 60°$$

argument = - 120° or 240°

Sol.25. (b)

Since, $\arg(z) - \arg(w) = \frac{\pi}{2}$

$$\Rightarrow -\arg(z) + \arg(w) = -\frac{\pi}{2}$$

$$\Rightarrow \arg(\bar{z}) + \arg(w) = -\frac{\pi}{2}$$

$$\Rightarrow \arg(\bar{z}w) = -\frac{\pi}{2} \quad ....(i)$$

Also given, $|zw| = i = |\bar{z}w| \quad ....(ii)$
$\therefore$ From Eqs. (i) and (ii), we get
$\bar{z}w = |zw|\, e^{i\arg(zw)}$

$$= 1\, e^{-i\frac{\pi}{2}} = \cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)$$

$$= 0 - i\sin\frac{\pi}{2} = -i$$

Sol.26. (b)

We have, $|z+4| \le 3 \quad \Rightarrow -3 \le z+4 \le 3$
$\Rightarrow -6 \le z+1 \le 0 \qquad \Rightarrow 0 \le (z+1) \le 6$
$\Rightarrow 0 \le |z+1| \le 6$
Hence, greatest and least values of $|z + 1|$ are 6 and 0, respectively

Sol.27. (a)

We know that, $\quad |-z| = |z|$
and $|z_1 + z_2| \le |z_2|$
Now, $|z| + |z-1| = |z| + |1+z| \ge |z + (1-z)| = |1| = 1$
Hence, minimum value of $|z| + |z-1|$ is 1

Sol.28. (b)

Since, $|z + a| \le a$ implies z lies on or inside a circle with centre $(-a, 0)$ and radius a, we have $|z_1| + |z_2| + |z_3| \le 4$

Sol.29. (b)

$$\left|z + \frac{2}{z}\right| = 2 \quad \Rightarrow |z| - \frac{2}{|z|} \le 2$$

$\Rightarrow |z|^2 - 2|z| - 2 \le 0$
This is a quadratic equation in $|z|$

$$\therefore |z| \le \frac{2 \pm \sqrt{4+8}}{2} \le 1 \pm \sqrt{3}$$

Hence, maximum value of $|z|$ is $1+\sqrt{3}$

Sol.30. (d)

Z =

$$\left(\frac{1+2i}{2+i}\right)^2 = \frac{1-4+4i}{4-1+4i} = \frac{3+4i}{3+4i} \times \frac{3-4i}{3-4i}$$

$$= \frac{-9+12i+12i+16}{9+16}$$

$$= \frac{7}{25} + \frac{24}{25}i$$

$$\bar{Z} = \frac{7}{25} - \frac{24}{25}i$$

Sol.31. (b)

$1+i^2+i^4+i^6+\ldots\ldots+i^{100}$
$= 1 - 1 + 1 - 1 + \ldots\ldots + 1 = 1$

Sol.32. (b)

$$\omega^{10} + \omega^{-10} = \omega^{10} + \frac{1}{\omega^{10}}$$

$$= \omega + \frac{1}{\omega} = \omega + \omega^2 = -1$$

Sol.33. (c)

$$\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right) = \frac{\left(\sqrt{3}+i\right)\left(\sqrt{3}+i\right)}{\left(\sqrt{3}-i\right)\left(\sqrt{3}+i\right)}$$

$$= \frac{\left(\sqrt{3}+i\right)^2}{3-i^2} = \frac{3+i^2+2\sqrt{3}i}{4} = \frac{2+2\sqrt{3}i}{4}$$

$$= \frac{1+\sqrt{3}i}{2} = -\omega^2$$

Now, $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^6 = (-\omega^2)^6 = \omega^{12} = 1$

Sol.34. (a)

$$\frac{1+2i}{1-(1-i)^2} = \frac{1+2i}{1-(1-1-2i)} = \frac{1+2i}{1+2i} = 1$$

$$\therefore \left|\frac{1+2i}{1-(1-i)^2}\right| = 1$$

Sol.35. (b)

Now, $\frac{i+\sqrt{3}}{-i+\sqrt{3}} = \frac{\left(i+\sqrt{3}\right)^2}{\left(\sqrt{3}-i\right)\left(\sqrt{3}+i\right)}$

$$= \frac{i^2+3+2\sqrt{3}i}{3+1} = \frac{-1+3+2\sqrt{3}i}{4}$$

$$= \frac{1+\sqrt{3}i}{2} = -\omega^2$$

and $\frac{i-\sqrt{3}}{i+\sqrt{3}} = \frac{\left(i-\sqrt{3}\right)^2}{i^2-\left(\sqrt{3}\right)^2}$

$$= \frac{i^2+3+2\sqrt{3}i}{-4} = \frac{2-2i\sqrt{3}}{-4} = \frac{-1+i\sqrt{3}}{2} = \omega$$

$$\therefore \left(\frac{i+\sqrt{3}}{-i+\sqrt{3}}\right)^{200} + \left(\frac{i-\sqrt{3}}{i+\sqrt{3}}\right)^{200} + 1$$

$= (-\omega^2)^{200} + \omega^{200} + 1$
$= \omega^{3\times133+1} + \omega^{3\times66+2} + 1$
$= \omega + \omega^2 + 1 = 0$

Sol.36. (b)

$$\frac{1+x+iy}{1+x-iy} = \frac{(1+x+iy)(1+x+iy)}{(1+x-iy)(1+x+iy)}$$

$$= \frac{(1+x)^2 + iy(1+x) + iy(1+x) - y^2}{1+x^2+2x+y^2}$$

$=\frac{1+x^2+2x-y^2+2iy(1+x)}{2(1+x)}$

$=\frac{1-y^2+2x+x^2+2iy(1+x)}{2(1+x)}$

$=\frac{2x^2+2x+2iy(1+x)}{2(1+x)}$ $(\because x^2+y^2=1)$

$= x + iy$

Sol.37. (d)
Correct d option to 1

$\left|\frac{1+2i}{1-(1-1+2i)}\right|=\left|\frac{1+2i}{1-2i}\right|=\frac{\sqrt{5}}{\sqrt{5}}=1$

Sol.38. (c)

$\left(-\sqrt{-1}\right)^{4n+3}+\left(i^{41}+i^{-257}\right)^9$

$=(-i)^{4n+3}+\left(i+\frac{1}{i}\right)^9$ $(\because i^4=1)$

$= (-1)^{4n+3}\,(i)^{4n}\,(i)^3 + (i-i)^9$

$= -(1)(-i) + 0 = i$

Sol.39. (a)
Since, α is a complex root of unity such that
$\alpha^2+\alpha+1=0$
$\Rightarrow \alpha = \omega$ or ω^2
$\therefore \alpha^{31} = (\omega)^{31} = \omega = \alpha$

Sol.40. (a)
Let $z = 2\omega^2 + 3i$

$\frac{2\left(-1-\sqrt{3}\right)}{2}+3i$

$= -1 + \left(3-\sqrt{3}\right)i$

$\therefore\ \bar{z} = -1 - \left(3-\sqrt{3}\right)i$

$= -1 + \sqrt{3i}\ -3i = 2\omega - 3i$

Sol.41. (d)

$\frac{\sqrt{3}+i}{1+\sqrt{3}i}\times\frac{1-\sqrt{3}i}{1-\sqrt{3}i}$

$=\frac{\sqrt{3}+\sqrt{3}+i-3i}{1+3}=\frac{2\sqrt{3}-2i}{4}=\frac{\sqrt{3}}{2}-\frac{i}{2}$

Sol.42. (a)
$2x = 3 + 5i$
$\Rightarrow 2x - 3 = 5i$
$\Rightarrow (2x-3)^2 = (5i)^2$
$\Rightarrow 4x^2 - 12x + 9 = -25$
$\Rightarrow 4x^2 - 12x + 34 = 0$

$$\begin{array}{r|l} 4x^2-12+34 & 2x^3+2x^2-7x+72 \quad \left(\frac{1}{2}x+2\right) \\ & \underline{2x^3-6x^2+17x} \\ & \quad 8x^2-24x+72 \\ & \quad \underline{8x^2-24x+68} \\ & \qquad 4 \end{array}$$

$\Rightarrow 2x^3 + 2x^2 - 7x + 72$

$=\underbrace{\left(4x^2-12x+34\right)}_{\text{Zero}}\left(\frac{1}{2}x+2\right)+4$

$\Rightarrow 4$

Sol.43. (c)
Sol.44. (c)
Sol.45. (b)

$\Rightarrow 1-\frac{1}{1+\omega}-\frac{1}{1+\omega^2}$

$\Rightarrow 1-\frac{1}{\left(-\omega^2\right)}-\frac{1}{(-\omega)}$

$\Rightarrow 1-(-\omega)-(-\omega^2)$
$= 1 + \omega + \omega^2 = 0$

Sol.46. (a)
Let $a = 1, b = 2$
$a + ib \quad 1 + 2i$

$\frac{1}{-a+ib}=\frac{1}{-1+2i}\times\frac{1-2i}{-1-2i}$

$=\frac{-1-2i}{1+4}=\frac{-1}{5}-\frac{5}{5}i$

Equation of line passes through (1, 2) and $\left(\frac{-1}{5},\frac{-2}{5}\right)$ is

$y-2=\frac{\frac{-2}{5}-2}{\frac{-1}{5}-1}(x-1)$

$y - 2 = 2(x-1)$
$y - 2 = 2x - 2$
$2x - y = 0$
$2x = y$
Passes through origin

Sol.47. (d)
Let $Z = a + ib \quad w = x + iy$
$|z + w| = |z - w|$
$=|a + ib + x + iy| = |a + ib - x - iy|$
$=|(a+x) + i(b+y)| = |(a-x) + i(b-y)|$
$=$

$\sqrt{(a+x)^2+(b+y)^2}=\sqrt{(a-x)^2+(b-y)^2}$

$= a^2 + x^2 + 2ax + b^2 + y^2 + 2by$
$= a^2 + x^2 - 2ax + b^2 + y^2 - 2by$
$4ax + 4by = 0$
$ax + by = 0$
$Z\bar{w} = (a+ib)(x-iy) = ax + by + i(bx - ay)$
$= i(bx - ay)$ is purely imaginary.

Sol.48. (b)

$z=\frac{1-\sqrt{3}i}{2}=-\omega$

$\sqrt{z}=\sqrt{-\omega}=\sqrt{-1}\sqrt{\omega}=$

$i\left(\frac{-1-\sqrt{3}i}{2}\right)$

$=\left[\frac{-1}{2}i+\frac{\sqrt{3}}{2}\right]$

$\left(\frac{\sqrt{3}}{2}-\frac{1}{2}i\right)$

Sol.49. (d)
Sol.50. (c)

$\frac{a+b\omega+c\omega^2}{a\omega^2+b+c\omega}\times\frac{\omega}{\omega}=\left(\frac{a+b\omega+c\omega^2}{a+b\omega+c\omega^2}\right)\times\omega=\omega$

NDA PYQ

1. If $z = 1 + \cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$, then what is $|z|$ equal to?
(a) $2\cos \pi/5$ (b) $2\sin \pi/5$
(c) $2\cos \pi/10$ (d) $2\sin \pi/10$
[NDA (I) - 2011]

2. What is the modulus of $\frac{1}{1+3i} - \frac{1}{1-3i}$?
(a) 3/5 (b) 9/25
(c) 3/25 (d) 5/3
[NDA (I) - 2011]

3. If ω is the imaginary cube root of unity, then what is $(2 - \omega + 2\omega^2)^{27}$ equal to?
(a) $3^{27}\omega$ (b) $-3^{27}\omega^2$
(c) 3^{27} (d) -3^{27}
[NDA (I) - 2011]

4. What are the square roots of $-2i$? ($i = \sqrt{1}$)
(a) $\pm(1 + i)$ (b) $\pm(1 - i)$
(c) ±1 (d) ±1
[NDA (II) - 2011]

5. If $z = \frac{1+2i}{2-i} - \frac{2-i}{1+2i}$, then what is the value of $z^2 + z\bar{z}$
(a) 0 (b) –1
(c) 1 (d) 8
[NDA (II) - 2011]

6. The smallest positive integral value of n for which $\left(\frac{1-i}{1+i}\right)^n$ is purely imaginary with positive imaginary part, is:
(a) 1 (b) 3
(c) 4 (d) 5
[NDA (II) - 2011]

7. If $z = 1 + i\tan\alpha$ where $\pi < \alpha < \frac{3\pi}{2}$, then what is $|z|$ equal to?
(a) $\sec\alpha$ (b) $-\sec\alpha$
(c) $\sec^2\alpha$ (d) $-\sec^2\alpha$
[NDA (II) - 2011]

8. What is the argument of $(1 - \sin\theta) + i\cos\theta$?
(a) $\frac{\pi}{2} - \frac{\theta}{2}$ (b) $\frac{\pi}{2} + \frac{\theta}{2}$
(c) $\frac{\pi}{4} - \frac{\theta}{2}$ (d) $\frac{\pi}{4} + \frac{\theta}{2}$
[NDA (II) - 2011]

9. What is the value of $(1 + i)^5 + (1 - i)^5$ where $I = \sqrt{-1}$?
(a) –8 (b) 8
(c) 8i (d) –8i
[NDA-2011(2)]

10. If α and β are the complex cube roots of unity, then what is the value of $(1 + \alpha)(1 + \beta)(1 + \alpha^2)(1 + \beta^2)$?
(a) –1 (b) 0
(c) 1 (d) 4
[NDA-2011(2)]

11. Let α and β be the roots of the equation $x^2+x+1=0$. The equation whose roots are α^{19} and β^7 is:
(a) $x^2 - x - 1 = 0$ (b) $x^2 - x + 1 = 0$
(c) $x^2 + x - 1 = 0$ (d) $x^2 + x + 1 = 0$

[NDA-2011(2)]

12. If p, q, r are positive integers and ω is the cube root of unity and $f(x) = x^{3p} + x^{3q+1} + x^{3r+2}$, then what is $f(\omega)$ equal to?
(a) ω (b) $-\omega^2$
(c) $-\omega$ (d) 0
[NDA-2011(2)]

13. If $A + iB = \frac{4+2i}{1-2i}$ where $i = \sqrt{-1}$, then what is the value of A?
(a) –8 (b) 0
(c) 4 (d) 8
[NDA (I) - 2012]

14. If $z = -\bar{z}$, then which one of the following is correct
(a) The real part of z is zero
(b) The imaginary part of z is zero
(c) The real part of z is equal to imaginary of part
(d) The sum of real and imaginary parts of z is z
[NDA (I) - 2012]

15. What is the modulus of $\frac{\sqrt{2}+i}{\sqrt{2}-i}$, where $i = \sqrt{-1}$?
(a) 3 (b) 1/2
(c) 1 (d) None of these
[NDA (II) - 2012]

16. Consider the following statements
I. $(\omega^{10} + 1)^7 + \omega = 0$
II. $(\omega^{105} + 1)^{10} = p^{10}$ for some prime number p,
Where, $\omega \neq 1$ is a cubic root of unity.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) neither I nor II
[NDA (II) - 2012]

17. The value of the sum $\sum_{n=1}^{13}(i^n + i^{n+1})$, where $i = \sqrt{-1}$ is
(a) i (b) –I
(c) 0 (d) i – 1
[NDA (II) - 2012]

18. What is the value of $\sqrt{-i}$, where $i = \sqrt{-1}$?
(a) $\pm\frac{1-i}{\sqrt{2}}$ (b) $\pm\frac{1+i}{\sqrt{2}}$
(c) $\pm\frac{1-i}{2}$ (d) $\pm\frac{1+i}{2}$
[NDA (I) - 2013]

19. What is the argument of the complex number $-1 - i$, where $i = \sqrt{-1}$?
(a) $\frac{5\pi}{4}$ (b) $-\frac{5\pi}{4}$
(c) $\frac{3\pi}{4}$ (d) None of these
[NDA (I) - 2013]

20. What is one of the square roots of $3 + 4i$, where $i = \sqrt{-1}$?
(a) 2 + i (b) 2 – I
(c) –2 + i (d) –3 – i
[NDA (II) - 2013]

21. What is the argument of the complex number $\frac{(1+i)(2+i)}{3-i}$, where $i=\sqrt{-1}$?
(a) 0 (b) $\pi/4$
(c) $-\pi/4$ (d) $\pi/2$
[NDA (I) - 2014]

22. If P and Q are two complex numbers, then the modulus of the quotient of P and Q is
(a)Greater than the quotient of their moduli
(b)Less than the quotient of their moduli
(c)Less than or equal to the quotient of their moduli
(d) Equal to the quotient of their moduli
[NDA (I) - 2014]

23. Let z = x + iy, where x, y are real variable and $i=\sqrt{-1}$. If $|2z-1| = |z-2|$, then the point z describes
(a) A circle (b) An ellipse
(c) A hyperbola (d) A parabola
[NDA (I) - 2014]

24. If $|z+\bar{z}|=|z-\bar{z}|$, then the locus of z is
(a) A pair of straight line
(b) A line
(c) A set of four straight lines
(d) A circle
[NDA (I) - 2014]

25. What is $\frac{(1+i)^{4n+5}}{(1-i)^{4n+3}}$ equal to, where n is a natural number and $i=\sqrt{-1}$?
(a) 2 (b) 2i
(c) –2i (d) i
[NDA (II) - 2014]

26. What is $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^6$ equal to, where $i=\sqrt{-1}$?
(a) 1 (b) 1/6
(c) 6 (d) 2
[NDA (II) - 2014]

27. If z is a complex number such that $|z| = 4$ and $\arg(z) = \frac{5\pi}{6}$, then what is z equal to?
(a) $2\sqrt{3}+2i$ (b) $2\sqrt{3}-2i$
(c) $2\sqrt{3}-2i$ (d) $-\sqrt{3}+i$
[NDA (II) - 2014]

28. If 1, ω and ω^2 are the cube roots of unity, then the value of $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)$ is
(a) –1 (b) 0
(c) 1 (d) 2
[NDA (I) - 2015]

29. If $z=\frac{-2(1+2i)}{(3+i)}$, where $i=\sqrt{-1}$, then the argument θ ($-\pi < \theta \le \pi$) of z is
(a) $\frac{3\pi}{4}$ (b) $\frac{\pi}{4}$
(c) $\frac{5\pi}{6}$ (d) $-\frac{3\pi}{4}$
[NDA (I)- 2015]

30. What is the square root of i, where $i=\sqrt{-1}$?
(a) $\frac{1+i}{2}$ (b) $\frac{1-i}{2}$
(c) $\frac{1+i}{\sqrt{2}}$ (d) None of these
[NDA (I) - 2015]

31. What is the real part of $(\sin x + i\cos x)^3$, where $i=\sqrt{-1}$?
(a) –cos3x (b) –sin3x
(c) sin3x (d) cos3x
[NDA (I)- 2015]

32. (x^3-1) can be factorized as
(a) $(x-1)(x-\omega)(x+\omega^2)$
(b) $(x-1)(x-\omega)(x-\omega^2)$
(c) $(x-1)(x+\omega)(x+\omega^2)$
(d) $(x-1)(x+\omega)(x-\omega^2)$
Where ω is one of the cube roots of unity
[NDA (I)- 2015]

33. What is $\left(\frac{\sin\frac{\pi}{6}+i\left(1-\cos\frac{\pi}{6}\right)}{\sin\frac{\pi}{6}-i\left(1-\cos\frac{\pi}{6}\right)}\right)^3$ is equal to?
(a) 1 (b) –1
(c) i (d) –i
[NDA (I) - 2015]

34. If z_1 and z_2 are complex numbers with $|z_1| = |z_2|$, then which of the following is/are correct?
I. $z_1 = z_2$
II. real part of z_1 = real part of z_2
III. Imaginary part of z_1 = Imaginary part of z_2
Select the correct answer using the code given below
(a) Only I (b) Only II
(c) Only III (d) None of these
[NDA (II) - 2015]

35. If the point $z_1 = 1 + i$, where $i=\sqrt{-1}$, is the reflection of a point $z_2 = x + iy$ in the line $i\bar{z}-iz=5$, then the point z_2 is
(a) 1 + 4i (b) 4 + I
(c) 1 – i (d) –1 – i
[NDA (II) - 2015]

36. $z\bar{z}+(3-i)z+(3+i)\bar{z}+1=0$ represents a circle with
(a) Center (–3, –1) and radius 3
(b) Center (–3, 1) and radius 3
(c) Center (–3, –1) and radius 4
(d) Center (–3, 1) and radius 4
[NDA (II) - 2015]

Directions (for next two) Let z_1, z_2 and z_3 be non zero complex numbers satisfying $z^2 = i\bar{z}$, where $i=\sqrt{-1}$.

37. What is $z_1 + z_2 + z_3$ equal to?
(a) i (b) –I
(c) 0 (d) 1
[NDA (I) - 2016]

38. Consider the following statements
I. $z_1z_2z_3$ is purely imaginary
II. $z_1z_2 + z_2z_3 + z_3z_1$ is purely real.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I) - 2016]

Directions (for next two) Let Z be a complex number satisfying $\left|\frac{z-4}{z-8}\right|=1$ and $\left|\frac{z}{z-2}\right|=\frac{3}{2}$

39. What is $|z|$ equal to?

(a) 6 (b) 12
(c) 18 (d) 36

[NDA (I) - 2016]

40. What is $\left|\frac{z-6}{z+6}\right|$ equal to?

(a)3 (b) 2
(c) 1 (d) 0

[NDA (I)- 2016]

41. Suppose ω is a cube root of unity with $\omega \neq 1$. Suppose, P and Q are the points on the complex plane defined by ω and ω^2. If O is the origin, then what is the angle between OP and OQ?

(a) 60° (b) 90°
(c) 120° (d) 150°

[NDA (I)- 2016]

42. Suppose ω_1 and ω_2 are two distinct cube roots of unity different from 1. Then, what is $(\omega_1 - \omega_2)^2$ equal to?

(a) 3 (b) 1
(c) –1 (d) –3

[NDA (I) - 2016]

43. If $z = x + iy = \left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right)^{-25}$, where $i = \sqrt{-1}$, then what is the fundamental amplitude of $\frac{z-\sqrt{2}}{z-i\sqrt{2}}$?

(a) π (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$

[NDA (I) - 2016]

44. If $\mathrm{Re}\left(\frac{z-1}{z+1}\right) = 0$ where z = x + iy, is a complex number, then which one of the following is correct?

(a) $z = 1 + i$ (b) $|z| = 2$
(c) $z = 1 - i$ (d) $|z| = 1$

[NDA (II) - 2016]

45. The value of $\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^{107} + \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^{107}$ then what is the imaginary part of z equal to?

(a) 0 (b) 1/2
(c) 2 (d) 1

[NDA (II) - 2016]

46. What is the number of distinct solutions of the equation $z^2 + |z| = 0$ (where z is a complex number)?

(a) One (b) Two
(c) Three (d) five

[NDA (II) - 2016]

47. What is $\omega^{100} + \omega^{200} + \omega^{300}$ equal to, where ω is the cube root of unity?

(a) 1 (b) 3ω
(c) $3\omega^2$ (d) 0

[NDA (II) - 2016]

48. What is $\sqrt{\frac{1+\omega^2}{1+\omega}}$ equal to, where ω is the cube root of unity?

(a) 1 (b) ω
(c) ω^2 (d) iω, where $i=\sqrt{-1}$

[NDA-2016(2)]

49. The value of $i^{2n} + i^{2n+1} + i^{2n+2} + i^{2n+3}$, where $i = \sqrt{-1}$, is:

(a) 0 (b) 1
(c) i (d) –i

[NDA (I)- 2017]

50. The value of $\left(\frac{-1+i\sqrt{3}}{2}\right)^n + \left(\frac{-1-i\sqrt{3}}{2}\right)^n$ where n is not a multiple of 3 and $i = \sqrt{-1}$, is:

(a) 1 (b) –1
(c) i (d) –i

[NDA (I)- 2017]

51. If 1, ω, ω^2 are the cube roots of unity, then $(1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega + \omega^2)$ is equal to:

(a) –2 (b) –1
(c) 0 (d) 2

[NDA (I)- 2017]

52. The modulus and principal argument of the complex number $\frac{1+2i}{1-(1-i)^2}$ are respectively:

(a) 1, 0 (b) 1, 1
(c) 2, 0 (d) 2, 1

[NDA (I)- 2017]

53. If $|z + 4| \leq 3$, then the maximum value of $|z + 1|$ is:

(a) 0 (b) 4
(c) 6 (d)10

[NDA (I)- 2017]

54. The number of roots of the equation $z^2 = 2\overline{z}$ is:

(a) 2 (b) 3
(c) 4 (d) Zero

[NDA (I)- 2017]

55. The smallest positive integer n for which $\left(\frac{1+i}{1-i}\right)^n = 1$, is:

(a) 1 (b) 4
(c) 8 (d) 16

[NDA (II) - 2017]

56. If $\left|z - \frac{4}{z}\right| = 2$, then the maximum value of |z| is equal to:

(a) $1+\sqrt{3}$ (b) $1+\sqrt{5}$
(c) $1-\sqrt{5}$ (d) $\sqrt{5}-1$

[NDA (II)- 2017]

57. If |a| denotes the absolute value of an integer, then which of the following are correct?

1. $|ab| = |a||b|$
2. $|a+b| \leq |a| + |b|$
3. $|a-b| \geq ||a|-|b||$

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2017(2)]

58. What is the principle argument of (–1 –i) where $i = \sqrt{-1}$.

(a) π/4 (b) –π/4
(c) –3π/4 (d)3π/4

[NDA (I) - 2018]

59. Let α and β be real number and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct non-real roots with Re(z) = 1, then it is necessary that

(a) $\beta \in (-1, 0)$ (b) $|\beta| = 1$
(c) $\beta \in (1, \infty)$ (d) $\beta \in (0, 1)$

[NDA (I) - 2018]

60. The number of non-zero integral solution of the equation $|1 - 2i|^x = 5^x$ is
(a) Zero (b) One
(c) Two (d) Three

[NDA (I) - 2018]

61. If α and β are different complex number with $|\alpha| = 1$ than what is $\left|\frac{\alpha-\beta}{1-\alpha\bar{\beta}}\right|$ equal to?
(a) $|\beta|$ (b) 2
(c) 1 (d) 0

[NDA (I) - 2018]

62. What is $i^{1000} + i^{1001} + i^{1002} + i^{1003}$ is equal to (where $i = \sqrt{-1}$)?
(a) 0 (b) I
(c) –I (d) 1

[NDA (I) - 2018]

63. The modulus-amplitude form of $\sqrt{3}+i$, where $i = \sqrt{-1}$ is
(a) $2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)$ (b) $2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$
(c) $4\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)$ (d) $4\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$

[NDA (I) - 2018]

64. What is the value of the sum $\sum_{n=2}^{11}(i^n+i^{n+1})$, where $i = \sqrt{-1}$?
(a) i (b) 2i
(c) –2i (d) 1 + i

[NDA (I) - 2018]

65. What is the value of :
$\left(\frac{-1+i\sqrt{3}}{2}\right)^{3n}+\left(\frac{-1-i\sqrt{3}}{2}\right)^{3n}$
Where, $i = \sqrt{-1}$?
(a) 3 (b) 2
(c) 1 (d) 1

[NDA (II) - 2018]

66. Which one of the following is correct in respect of the cube roots of unity?
(a) They are collinear
(b) They lie on a circle of radius $\sqrt{3}$
(c) They form an equilateral triangle
(d) None of these

[NDA (II) - 2018]

67. If A={x∈Z:x^3–1=0} and B={x∈Z : $x^2 + x + 1 = 0$}, where Z is set of complex numbers, then what is A∩B equal to?
(a) Null set (b) $\left\{\frac{-1+\sqrt{3}i}{2},\frac{-1-\sqrt{3}i}{2}\right\}$
(c) $\left\{\frac{-1+\sqrt{3}i}{4},\frac{-1-\sqrt{3}i}{4}\right\}$ (d) $\left\{\frac{-1+\sqrt{3}i}{2},\frac{1-\sqrt{3}i}{2}\right\}$

[NDA-2019(1)]

68. The common roots of the equations $z^3 + 2z^2 + 2z + 1 = 0$ and $z^{2017} + z^{2018} + 1 = 0$ are:
(a) $-1, \omega$ (b) $1, \omega^2$
(c) $-1, \omega^2$ (d) ω, ω^2

[NDA-2019(1)]

Direction (For next two) A complex number is given as below $\frac{1+2i}{1-(1-i)^2}$

69. Modulus of z is
(a) 4 (b) 2
(c) 1 (d) 1/2

[NDA (I) - 2019]

70. Argument of z is
(a) 0 (b) $\pi/4$
(c) $\pi/2$ (d) π

[NDA (I) - 2019]

71. What is the value of $\left[\frac{i+\sqrt{3}}{2}\right]^{2019}+\left[\frac{i-\sqrt{3}}{2}\right]^{2019}=$
(a) 1 (b) – 1
(c) 2i (d) –2i

[NDA (II) - 2019]

72. If α and β are the roots of $x^2 + x + 1 = 0$, then what is the value of $\sum_{j=0}^{3}\left(\alpha^j+\beta^j\right)=$
(a) 8 (b) 6
(c) 4 (d) 2

[NDA (II) - 2019]

73. If x = 1 + i, then what is the value of $x^6 + x^4 + x^2 + 1$?
(a) 6i – 3 (b) –6i +3
(c) –6i – 3 (d) 6i +3

[NDA (II) - 2019]

74. What is the modulus of complex number $\frac{\cos\theta+i\sin\theta}{\cos\theta-i\sin\theta}$, where $i = \sqrt{-1}$
(a) 1/2 (b) 1
(c) 3/2 (d) 2

[NDA 2020]

75. What is the argument of complex number $\frac{1-i\sqrt{3}}{1+i\sqrt{3}}$, *where* $i=\sqrt{-1}$?
(a) 240° (b) 210°
(c) 120° (d) 60°

[NDA 2020]

76. Consider the following statements
I. $\left(\overline{z^{-1}}\right)=\left(\bar{z}\right)^{-1}$ II. $z\bar{z}^{-1}=|z|^2$
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) neither I nor II

[NDA (I) - 2021]

77. Consider the following statements
I.The difference of z and its conjugate is an imaginary number.
II.The sum of z and its conjugate is a real number.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) neither I nor II

[NDA (I) - 2021]

78. What is the modulus of complex number $i^{2n+1}(-i)^{2n-1}$, where n ∈ N and $i = \sqrt{-1}$?
(a) –1 (b) 1
(c) $\sqrt{2}$ (d) 2

[NDA (I) - 2021]

79. The smallest positive integer n for which

$\left(\frac{1-i}{1+i}\right)^{n^2}=1$, is

(a) 2 (b) 4
(c) 6 (d) 8

[NDA (I) - 2021]

80. If z = 1 + i ,then what is the modulus of $z+\frac{2}{z}$?

(a) 1 (b) 2
(c) 3 (d) 4

[NDA (I) - 2021]

81. If k is one root of the equation x (x+1) + 1 = 0 then what is other root?

(a) 1 (b) –k
(c) k^2 (d) $-k^2$

[NDA (I) - 2021]

82. Consider the following statements in respect of the roots of the equation $x^3-8=0$.

1.the roots are non-collinear
2.The roots lie on a circle of unit radius
Which of the above statement is/are correct?

(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2021]

83. What is $\sum_{n=1}^{8n+7} i^n$ equal to, where i=$\sqrt{-1}$?

(a) –1 (b) 1
(c) i (d) –i

[NDA (II) - 2021]

84. If z = x + iy, where i=$\sqrt{-1}$, then what does the equation $z\bar{z}+|z|^2+4(z+\bar{z})-48=0$ represent?

(a) Straight line (b) Parabola
(c) Circle (d) Pair of straight lines

[NDA (II) - 2021]

85. Which one of the following is a square root of 2a + $2\sqrt{a^2+b^2}$, where a, b∈R ?

(a) $\sqrt{a+ib}+\sqrt{a-ib}$ (b) $\sqrt{a+ib}-\sqrt{a-ib}$
(c) 2a + ib (d) 2a – ib
Where i=$\sqrt{-1}$

[NDA (II) - 2021]

86. If i=$\sqrt{-1}$, then how many values does i^{-2n} have for different n∈Z?

(a) One (b) Two
(c) Four (d) Infinite

[NDA (II) - 2021]

87. If $x^2+x+1=0$, then what is the value of $x^{199}+x^{200}+x^{201}$?

(a) –1 (b) 0
(c) 1 (d) 3

[NDA (II) 2021]

88. What is the principal argument of $\frac{1}{1+i}$ where i = $\sqrt{-1}$?

(a) $-\frac{3\pi}{4}$ (b) $-\frac{\pi}{4}$
(c) $\frac{\pi}{4}$ (d) $\frac{3\pi}{4}$

[NDA (I) - 2022]

89. What is the modules of $\left(\frac{\sqrt{-3}}{2}-\frac{1}{2}\right)^{200}$?

(a) 1/4 (b) 1/2
(c) 1 (d) 2^{200}

[NDA (I) - 2022]

Consider the following for the next three (03) items that follow:

Let z=$\frac{1+i\sin\theta}{1-i\sin\theta}$ where i=$\sqrt{-1}$

90. What is the modulus of z?

(a) 1 (b) $\sqrt{2}$
(c) $1+\sin^2\theta$ (d) $\frac{1+\sin^2\theta}{1-\sin^2\theta}$

[NDA 2022 (II)]

91. What is angle θ such that z is purely real?

(a) $\frac{n\pi}{2}$ (b) $\frac{(2n+1)\pi}{2}$
(c) nπ (d) 2nπ only
Where n is an integer

[NDA 2022 (II)]

92. What is angle θ such that z is purely imaginary?

(a) $\frac{n\pi}{2}$ (b) $\frac{(2n+1)\pi}{2}$
(c) nπ (d) 2nπ
Where n is an integer

[NDA 2022 (II)]

93. If z is a complex number such that $\frac{z-1}{z+1}$ is purely imaginary, then what is |z| equal to?

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) 1 (d) 2

[NDA – 2023 (1)]

94. If z = $\frac{1+i\sqrt{3}}{1-i\sqrt{3}}$ where i = $\sqrt{-1}$, then what is the argument of z?

(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$
(c) $\frac{4\pi}{3}$ (d) $\frac{5\pi}{6}$

[NDA – 2023 (1)]

95. If ω is a non-real cube root of 1, then what is the value of $\left|\frac{1-\omega}{\omega+\omega^2}\right|$?

(a) √3 (b) √2
(c) 1 (d) $\frac{4}{\sqrt{3}}$

[NDA – 2023 (1)]

96. If α and β are the distinct roots of equation $x^2-x+1=0$, then what is the value of $\left|\frac{\alpha^{100}+\beta^{100}}{\alpha^{100}-\beta^{100}}\right|$?

(a) √3 (b) √2
(c) 1 (d) $\frac{1}{\sqrt{3}}$

[NDA – 2023 (1)]

97. If $z\bar{z}=|z+\bar{z}|$, where z = x + iy, i =√–1, then the locus of z is a pair of:

(a) straight lines (b) rectangular hyperbolas
(c) parabola (d) circles

[NDA-2023 (2)]

98. What is the value of $\sqrt{12+5i}+\sqrt{12-5i}$, where $i=\sqrt{-1}$.

(a) 24 (b) 25
(c) $5\sqrt{2}$ (d) $5(\sqrt{2}-1)$

[NDA-2023 (2)]

99. Which one of the following is a square root of $-\sqrt{-1}$?

(a) $1+i$ (b) $\frac{1-i}{\sqrt{2}}$
(c) $\frac{1+i}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}i$

[NDA-2023 (2)]

Consider the following for the next (03) items that follow:
Consider the equation-I : $z^3+2z^2+2z+1=0$
equation-II: $z^{1985}+z^{100}+1=0$

100. What are the roots of equation-I?

(a) $1, \omega, \omega^2$ (b) $-1, \omega, \omega^2$
(c) $1, -\omega, \omega^2$ (d) $-1, -\omega, -\omega^2$

[NDA-2023 (2)]

101. Which one of the following is a root of equation-II?

(a) -1 (b) $-\omega$
(c) $-\omega^2$ (d) ω

[NDA-2023 (2)]

102. What is the number of common roots of equation-I and equation-II?

(a) 0 (b) 1
(c) 2 (d) 3

[NDA-2023 (2)]

103. If $\omega \neq 1$ is a cube root of unity, then what are the solutions of $(z-100)^3+1000=0$?

(a) $10(1-\omega)$, $10(10-\omega^2)$, 100
(b) $10(10-\omega)$, $10(10-\omega^2)$, 90
(c) $10(1-\omega)$, $10(10-\omega^2)$, 1000
(d) $(1+\omega)$, $(10+\omega^2)$, -1

[NDA-2024 (1)]

104. What is $(1+i)^4+(1-i)^4$ equal to, where $i=\sqrt{-1}$?

(a) 4 (b) 0
(c) -4 (d) -8

[NDA-2024 (1)]

105. If z is any complex numbe rand $iz^3+z^2-z+i=0$, where $i=\sqrt{-1}$, then what is the value of $(|z|+1)^2$?

(a) 1 (b) 4
(c) 81 (d) 121

[NDA-2024 (1)]

106. If x, y and z are the cube roots of unity, then what is the value of $xy+yz+zx$?

(a) 0 (b) 1
(c) 2 (d) 3

[NDA-2024 (1)]

107. Let z_1 and z_2 be two complex numbers such that $\left|\frac{z_1+z_2}{z_1-z_2}\right|=1$, then what is $\text{Re}\left(\frac{z_1}{z_2}\right)+1$ equal to?

(a) -1 (b) 0
(c) 1 (d) 5

[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below
Let Z_1 and Z_2 be any two complex numbers such that $Z_1^2+Z_2^2+Z_1Z_2=0$.

108. What is the value of $\left|\frac{Z_1}{Z_2}\right|$?

(a) 1 (b) 2
(c) 3 (d) 4

[NDA-2024 (2)]

109. What is the value of $\frac{1}{2}+\text{Re}\left(\frac{Z_1}{Z_2}\right)$?

(a) -1 (b) 0
(c) 1 (d) 2

[NDA-2024 (2)]

110. If $\omega \neq 1$ is a cube root of unity, then what is $(1+\omega-\omega^2)^{100}+(1-\omega+\omega^2)^{100}$ equal to ?

(a) $2^{100}\omega^2$ (b) $2^{100}\omega$
(c) 2^{100} (d) -2^{100}

[NDA-2024 (2)]

111. If $z=\frac{1}{3}\begin{vmatrix} i & 2i & 1 \\ 2i & 3i & 2 \\ 3 & 1 & 3 \end{vmatrix}=x+iy; i=\sqrt{-1}$

then what is modulus z equal to?

(a) 1 (b) $\sqrt{2}$
(c) 2 (d) $\sqrt{3}$

[NDA-2024 (2)]

112. What is the value of the sum $\sum_{n=1}^{20}\left(i^{n-1}+i^n+i^{n+1}\right)$?

(a) $-2i$ (b) 0
(c) 1 (d) $2i$

[NDA-2024 (2)]

113. What is $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^3$ equal to?

(a) -1 (b) 0
(c) 1 (d) 3

[NDA-2025 (1)]

114. If $x^2-x+1=0$, then what is $\left(x-\frac{1}{x}\right)^2+\left(x-\frac{1}{x}\right)^4+\left(x-\frac{1}{x}\right)^8$ equal to?

(a) 81 (b) 85
(c) 87 (d) 90

[NDA-2025 (1)]

115. If $z \neq 0$ is a complex number, then what is $amp(z)+amp(\bar{z})$ equal to?

(a) 0 (b) $\pi/2$
(c) π (d) 2π

[NDA-2025 (1)]

116. If $\alpha=\frac{-1+\sqrt{-3}}{2}$, then what is the value of $(1+\alpha^{19}-\alpha^{35})^{100}-(1-3\alpha^{25}+\alpha^{38})^{50}$

(a) -2 (b) -1
(c) 0 (d) 2

[NDA-2025 (2)]

117. If $\left(\frac{1-i}{1+i}\right)^{2m}\left(\frac{1+i}{1-i}\right)^{2n}=1$, What is the smallest positive value of $m-n$?

(a) 1 (b) 2
(c) 4 (d) 8

[NDA-2025 (2)]

118. Let 1, ω, ω^2 be three cube roots of unity. If $x = a + b$, $y = a\omega + b\omega^2$, $z = a\omega^2 + b\omega$, then what is $x^2 + y^2 + z^2$ equal to?
(a) 6ab (b) 3ab (c) $a^2 + b^2$ (d) 1 **[NDA-2025 (2)]**

ANSWER KEY

1.	c	**2.**	a	**3.**	d	**4.**	b	**5.**	a	**6.**	b	**7.**	b	**8.**	d	**9.**	a	**10.**	c
11.	d	**12.**	d	**13.**	b	**14.**	a	**15.**	c	**16.**	b	**17.**	d	**18.**	a	**19.**	a	**20.**	a
21.	d	**22**	d	**23.**	a	**24.**	a	**25.**	a	**26.**	a	**27.**	a	**28.**	c	**29.**	d	**30.**	c
31.	b	**32.**	b	**33.**	c	**34.**	d	**35.**	a	**36.**	a	**37.**	c	**38.**	c	**39.**	a	**40.**	d
41.	c	**42.**	d	**43.**	a	**44.**	d	**45.**	a	**46.**	c	**47.**	d	**48.**	b	**49.**	a	**50.**	b
51.	c	**52.**	a	**53.**	c	**54.**	c	**55.**	b	**56.**	b	**57.**	d	**58.**	c	**59.**	c	**60.**	a
61.	c	**62.**	a	**63.**	b	**64.**	c	**65.**	b	**66.**	c	**67.**	b	**68.**	d	**69.**	c	**70.**	a
71.	c	**72.**	d	**73.**	c	**74.**	b	**75.**	a	**76.**	a	**77.**	c	**78.**	b	**79.**	a	**80.**	b
81.	c	**82.**	a	**83.**	a	**84.**	c	**85.**	a	**86.**	b	**87.**	b	**88.**	b	**89.**	c	**90.**	a
91.	c	**92.**	b	**93.**	c	**94.**	b	**95.**	a	**96.**	d	**97.**	d	**98.**	c	**99.**	b	**100.**	b
101.	d	**102.**	c	**103.**	b	**104.**	d	**105.**	b	**106.**	a	**107.**	c	**108.**	a	**109.**	b	**110.**	d
111.	b	**112.**	b	**113.**	a	**114.**	c	**115.**	a	**116.**	c	**117.**	b	**118.**	a				

Solutions

Sol. 1. (c)

$$z = 1 + \cos\frac{\pi}{5} + i\sin\frac{\pi}{5}$$

$$|z| = \sqrt{\left(\cos\frac{\pi}{5}+1\right)^2 + \left(\sin\frac{\pi}{5}\right)^2}$$

$$= \sqrt{1+\cos^2\frac{\pi}{5} + 2\cos\frac{\pi}{5} + \sin^2\frac{\pi}{5}}$$

$$= \sqrt{2+2\cos\frac{\pi}{5}}$$

$$= \sqrt{2}\left\{\sqrt{1+2\cos^2\frac{\pi}{10}-1}\right\} = 2\cos\frac{\pi}{10}$$

Sol. 2. (a)

$$\frac{1}{1+3i} - \frac{1}{1-3i} = \frac{1-3i-1-3i}{(1-9i^2)} \quad (\because i^2 = -1)$$

$$= \frac{6i}{10} = -\frac{3i}{5}$$

$$\therefore \text{Modulus} = \left|-\frac{3}{5}i\right| = \sqrt{0^2 + \left(\frac{-3}{5}\right)^2} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

Sol. 3. (d)

$(2-\omega+2\omega^2)^{27}$ (given, $\omega^3 = 1$ and $1+\omega^2 = 0$)

$= -(3)^{27}(\omega^3)^9$

$=[2(1+w^2)-w]^{27}$

$=(-2w-w)^{27}$

$=(-3w)^{27}$

$= (-3)^{27}(1)^9 = -(3)^{27}$

Sol. 4. (b)

Square root of $(-2i)$ i.e., $(-2i)^{1/2}$ …(i)

Let $z = r(\cos\theta + i\sin\theta) = 0 - 2i$

On comparing both sides, we getr $\cos\theta = 0$ …(ii)

$r\sin\theta = -2$ …(iii)

On squaring equation (ii) and (iii) and adding both equations, we get

$r^2 = 4 \Rightarrow r = \pm 2$

equation (iii) divide by equation (ii),

$$\tan\theta = -\infty = \tan\left(-\frac{\pi}{2}\right)$$

$$\theta = -\frac{\pi}{2}$$

From equation.(i),

$(-2i)^{1/2} = \{\pm 2\cos(-\pi/2) + i\sin(-\pi/2)\}^{1/2}$

$= \pm 2^{1/2}\{\cos(-\pi/2) + i\sin(-\pi/2)\}^{1/2}$

$= \sqrt{2}\{\cos \pi/4 - i\sin\pi/4\}$

(by De-moiver theorem)

$$= \pm\sqrt{2}\left\{\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right\}$$

$$= \pm\sqrt{2}\frac{(1-i)}{\sqrt{2}} = \pm(1-i)$$

Sol. 5. (a)

$$Z = \frac{1+2i}{2-i} - \frac{2-i}{1+2i}$$

$$= \frac{1-4+4i-4+1+4i}{(1+2i)(2-i)} = \frac{-6+8i}{2+2+3i} = \frac{-6+8i}{4+3i}$$

$$\Rightarrow \frac{-6+8i}{4+3i}\times\frac{4-3i}{4-3i} = \frac{-24+24+50i}{25} = 2i$$

$Z^2 + Z\bar{Z}$

$(2i)^2 + (2i)(-2i)$

$-4 - 4i = 0$

Sol. 6. (b)

$$\text{Let } z = \left(\frac{1-i}{1+i}\right)^n = \left\{\frac{(1-i)(1-i)}{(1+i)(1-i)}\right\}^n$$

$$= \left\{\frac{(1-i)^2}{1^2-i^2}\right\}^n = \left\{\frac{1+i^2-2i}{1+1}\right\}^n = \left\{\frac{1-1-2i}{2}\right\}^n$$

$=(-i)^n$

Here, the smallest positive integral value of n for which 'z' is purely imaginary with positive imaginary part should be 3.

$\Rightarrow (-i)^3 = -i^3 = -i^2.\, i = -(-1).\, i = i$

Sol. 7. (a)

Given, $z = 1 + i\tan\alpha$, where $\pi < \alpha <$ [illegible]

$|z| = |1 + i\tan\alpha| = |\sec\alpha|$

for $\pi < \alpha <$ [illegible] $\sec\alpha$ is negative, but modulus is always positive so answer is $-\sec\alpha$.

Sol. 8. (d)

Let $z = (1-\sin\theta) + i\cos\theta$

$$\Rightarrow \arg(z) = \tan^{-1}\left(\frac{\operatorname{Im}(z)}{\operatorname{Re}(z)}\right) = \tan^{-1}\left(\frac{\cos\theta}{1-\sin\theta}\right)$$

$$=\tan{-1}\left\{\frac{\left(\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}\right)}{\sin^2\frac{\theta}{2}+\cos^2\frac{\theta}{2}-2\sin\frac{\theta}{2}.\cos\frac{\theta}{2}}\right\}$$

$$=\tan{-1}\left\{\frac{\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right)\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)}{\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right)}\right\}$$

$$=\tan{-1}\left\{\frac{\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)}{\cos\frac{\theta}{2}-\sin\frac{\theta}{2}}\right\} = \tan^{-1}\left(\frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}}\right)$$

$$= \tan^{-1}\tan\left[\frac{\pi}{4}+\frac{\theta}{2}\right] = \frac{\pi}{4}+\frac{\pi}{2}$$

Sol. 9. (a)

$(1+i)^5 + (1-i)^5$

$=(1 + 5i + 10i^2 + 10i^3 + 5i^4 + i^5)$

$+ (1 - 5i + 10i^2 - 10i^3 + 5i^4 - i^5)$

$2 - 20 + 10 = -8$

Sol. 10. (c)

$(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)$

$(-\omega^2)(-\omega)(-\omega^2)(-\omega)=1$

Sol. 11. (d)

We know that roots of $x^2 + x + 1 = 0$ are ω and ω^2

$\omega^{19} = \omega$ and $(\omega^2)^7 = \omega^2$

So

required equation will be same as $x^2 + x + 1 = 0$

Sol. 12. (d)

$1 + \omega + \omega^2 = 0$

Sol. 13. (b)

$$\text{Given, } A + iB = \frac{4+2i}{1-2i}\times\frac{1+2i}{1+2i}$$

$$= \frac{4+2i+8i+4i^2}{1-4i^2} \quad (\because i^2 = -1)$$

$$= \frac{4+10i-4}{1+4} = \frac{10i}{5} = 2i$$

$\Rightarrow A + iB = 0 + i2$

$\Rightarrow A = 0$ and $B = 2$

Sol. 14. (a)

Given that, $z = -\bar{z}$

Let $z = x + iy$

$\Rightarrow (x+iy) = -(\overline{x+iy})$

$(x + iy) = -(x - iy)$

$[\because \quad y, \bar{z} = x = -iy]$

$\Rightarrow (x + iy) = (-x + iy)$

$\Rightarrow 2x = 0$

$\Rightarrow x = 0$

$\therefore z = x + iy = 0 + iy = iy$

Hence, the real part of z is zero.

Sol. 15. (c)

$$\frac{\sqrt{2}+i}{\sqrt{2}-i} = \frac{\sqrt{2}+i}{\sqrt{2}-i}\times\frac{\sqrt{2}+i}{\sqrt{2}+i}$$

$$= \frac{\left(\sqrt{2}+i\right)^2}{\left(\sqrt{2}\right)^2-(i)^2} = \frac{2+i^2+2\sqrt{2}i}{2-i^2}$$

$$= \frac{2-1+2\sqrt{2}i}{2-(-1)} = \frac{1+2\sqrt{2}i}{3}$$

$$\therefore \frac{\sqrt{2}+1}{\sqrt{2}-i} = \frac{1+2\sqrt{2}i}{3}$$

$$\Rightarrow \frac{\sqrt{2}+i}{\sqrt{2}-i} = \frac{1}{3}+\frac{2\sqrt{2}}{3}i$$

$$\Rightarrow \left|\frac{\sqrt{2}+1}{\sqrt{2}-i}\right| = \left|\frac{1}{3}+\frac{1+2\sqrt{2}}{3}i\right|$$

$$= \sqrt{\left(\frac{1}{3}\right)^2+\left(\frac{2\sqrt{2}}{3}\right)^2} = \sqrt{\frac{1}{9}+\frac{8}{9}} = \sqrt{\frac{9}{9}} = 1$$

Alternate method:

We know that,

If z_1 and z_2 are two complex numbers (by complex number properties).

$$\text{Then, } \left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|},$$

Provided $Z_2 \neq 0$

$$\therefore \left|\frac{\sqrt{2}+i}{\sqrt{2}-i}\right| = \frac{|\sqrt{2}+i|}{|\sqrt{2}-i|} = \frac{\sqrt{2+1}}{\sqrt{2+1}} = \frac{\sqrt{3}}{\sqrt{3}} = 1$$

Sol. 16. (b)

Statement I.

LHS $= (\omega^{10} + 1)^7 + \omega$

$= [(\omega^3)^3\,\omega + 1]^7 + \omega$

$= (1 + \omega)^7 + \omega$

$= (-\omega^2)^7 + \omega$

$(\because 1 + \omega + \omega^2 = 0, \therefore 1 + \omega = -\omega^2]$

$= -\omega^{14} + \omega = -(\omega^3)^4\,\omega^2 + \omega$
$= -\omega^2 + \omega = (1 + \omega) + \omega$
$= 1 + 2\omega \neq 0$
So, Statement I is false.
Statement II.
LHS $= (\omega^{105} + 1)^{10} = [(\omega^3)^{35} + 1]^{10}$ ($\because \omega^3 = 1$)
$= (1 + 1)^{10}$
$= 2^{10} = p^{10}$ which is true for prime number 2.
So, Statement I is false and Statement II is true.

Sol. 17. (d)
Reqd. sum $= \dfrac{i(1-i^{13})}{1-i} + \dfrac{i^2\left(1-i^{13}\right)}{1-i}$
[By G.P.]
$= i\left(\dfrac{1-i^{13}}{1-i}\right)\left(i+i^2\right) = \left(\dfrac{1-i}{1-i}\right)\left(i-1\right) = i-1$

Sol. 18. (a)
Let $z = -i = 0 - i = r(\cos\theta + i\sin\theta)$ …(i)
Here, by comparing real and imaginary parts on both sides,
We get
$r\cos\theta = 1$ …(ii)
and $r\sin\theta = -1$ ...(iii)
On squaring and adding Eqs. (ii) and (iii), we get
$r^2\sin^2\theta + r^2\cos^2\theta = 0 + 1$
$\Rightarrow r^2(\sin^2\theta + \cos^2\theta) = 1 \Rightarrow r^2 = 1$
$\therefore r = \pm 1$
On dividing equation (iii) by Equation (ii), we get
$\tan\theta = \infty = \tan 90° \Rightarrow \theta = 90° = \dfrac{\pi}{2}$
But principle argument of $z = -\theta = \dfrac{-\pi}{2}$
(since, z lies in IVth quadrant)
$\therefore z = -1 = 1\left\{\cos\left(\dfrac{-\pi}{2}\right) + i\sin\left(\dfrac{-\pi}{2}\right)\right\}$
$z = -1 = 1\left\{\cos\left(\dfrac{\pi}{2}\right) - i\sin\left(\dfrac{\pi}{2}\right)\right\}$
[$\because \cos(-\theta) = \cos\theta$]
Now, $\sqrt{z} = \sqrt{-i} = \pm(1)^{1/2}\left(\cos\dfrac{\pi}{2} - i\sin\dfrac{\pi}{2}\right)^{1/2}$

Sol. 19. (a)
Let $z = -1 - i = r(\cos\theta + i\sin\theta)$
On comparing both sides real and imaginary parts, we get
$r\cos\theta = -1$ …(i)
and $r\sin\theta = -1$ …(ii)
On dividing Equation. (ii) by Equation (i), we get
$\dfrac{r\sin\theta}{r\cos\theta} = \dfrac{-1}{-1} \Rightarrow \tan\theta = 1 = \tan\dfrac{\pi}{4}$
$\Rightarrow \theta = \dfrac{\pi}{4}$
Since, argument of z lies in the IIIrd quadrant
$\therefore \arg(z) = \pi + \theta = \pi + \dfrac{\pi}{4} = \dfrac{5\pi}{4}$

Sol. 20. (a)
Let $x + iy = \sqrt{3+4i}$
On squaring both sides
$(x + iy)^2 = 3 + 4i$
$\Rightarrow x^2 - y^2 + 2xyi = 3 + 4i$
On equating real and imaginary parts on both sides,
$x^2 - y^2 = 3$ ...(i)
and $2xy = 4$ …(ii)
Now, we use the following identity.
$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$
$\Rightarrow (x^2 + y^2)^2 = (3)^2 + (4)^2$
$= 9 + 16 = 25$
$\Rightarrow x^2 + y^2 = 5$ …(iii)
From Equation (i) and (ii),
$x^2 = 4$ and $y^2 = 1$
$\Rightarrow x = \pm 2$, and $y = \pm 1$
Since, the product of xy is spositive
$\therefore x = 2$, and $y = 1$
or $x = -2$, and $y = -1$
Thus, the square root of the complex number $3 + 4i$ is $\pm 2 \pm i$.

Sol. 21. (d)
Let $z = \dfrac{(1+i)(2+i)}{(3-i)}$
$= \dfrac{2+3i+i^2}{3-1} = \dfrac{2+3i-1}{3-i}$ [$\because$
$= \dfrac{1+3i}{3-i} \times \dfrac{3+i}{3+i} = \dfrac{3+10i+3i^2}{9-i}$
$= \dfrac{3+10i-3}{9+1} = \dfrac{10i}{10} = i$ [$\because$
$\therefore z = 0 + i.\,1$
So, $\arg(z) = \tan^{-1}\left(\dfrac{1}{0}\right) = \tan^{-1}(\infty) = \pi/2$

Sol. 22. (d)
Let $P = x + iy$ and $Q = \alpha + i\beta$ be two complex numbers.
Then, its quotient $= \dfrac{P}{Q} = \dfrac{x+iy}{\alpha+i\beta}$
Now, $\left|\dfrac{P}{Q}\right| = \left|\dfrac{x+iy}{\alpha+i\beta}\right|$
$= \sqrt{\dfrac{x^2+y^2}{\alpha^2+\beta^2}} = \sqrt{\dfrac{x^2+y^2}{\alpha^2+\beta^2}} = \dfrac{|P|}{|Q|}$
Hence, the modulus of the quotient of P and Q is equal to the quotient of their moduli i.e.,
$\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$

Sol. 23. (a)
Given that, $z = x + iy$; $x, y \in R$
We have, $|2z - 1| = |z - 2|$
$\Rightarrow |2(x + iy) - 1| = |x + iy - 2|$
$\Rightarrow |(2x - 1) + 2iy| = |(x - 2) + iy|$
$\Rightarrow |(2x - 1) + 2iy|^2 = |(x - 2) + iy|^2$
$\Rightarrow (2x - 1)^2 + (2y)^2 = (x - 2)^2 + y^2$
$\Rightarrow 4x^2 + 1 - 4x + 4y^2 = x^2 + 4 - 4x + y^2$
$\Rightarrow 3x^2 + 3y^2 - 3 = 0$
$\Rightarrow x^2 + y^2 = 1$
Which represents a circle.
Hence, point z describes a circle.

Sol. 24. (a)
Given condition, $|z + \bar{z}| = |z - \bar{z}|$
Let $z = x + iy \Rightarrow \bar{z} = x - iy$
$\therefore |x + iy + x - iy| = |x + iy - x + iy|$
$\Rightarrow |2x| = |2iy|$ [$\because$ $= \sqrt{a^2 + b^2}$]
$\Rightarrow 2x = 2y$
$x = y$
Which represent pair of straight lines passing through the origin. Hence, the locus of z is a pair of straight lines.

Sol. 25. (a)
We have,
$\dfrac{(1+i)^{4n+5}}{(1-i)^{4n+3}} = \dfrac{(1+i)^{4n+3}.(1+i)^2}{(1-i)^{4n+3}}$
$= \left(\dfrac{1+i}{1-i}\right)^{4n+3}.(1+i)^2$
$\left[\dfrac{(1+i)(1+i)}{(1-i)(1+i)}\right]^{4n+3}.\ (1+i^2+2i)$
$= \left[\dfrac{1+i^2+2i}{1+1}\right]^{4n+3}.\ 2i$
$= (i)^{4n+3}.\ 2i$
$= 2(i)^{4n+4} = 2$

Sol. 26. (a)
$\left(\dfrac{\sqrt{3}+i}{\sqrt{3}-i}\right) = \dfrac{\sqrt{3}+i}{\sqrt{3}-i} \times \dfrac{\sqrt{3}+i}{\sqrt{3}+i}$
$= \dfrac{3+i^2+2\sqrt{3}i}{3-i^2} = \dfrac{3-1+2\sqrt{3}i}{3+1}$
$= \dfrac{2(1+\sqrt{3}i)}{4} = \dfrac{1}{2} + i\dfrac{\sqrt{3}}{2}$
$= \left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right)$
$\therefore \left(\dfrac{\sqrt{3}+i}{\sqrt{3}-i}\right)^6 = \cos 2\pi + i\sin 2\pi = 1 + 0.i = 1$

Sol. 27. (a)
Let $z = a+ib$ and $|z| = a^2 + b^2 = 4$
Since, $\arg. z = \dfrac{5\pi}{6}$
So, z lies in second quadrant
$\Rightarrow \left(\pi - \dfrac{5\pi}{6}\right) = \dfrac{\pi}{6}$
$z = r(\cos\alpha + i\sin\alpha)$
$z = r\left(\cos\dfrac{\pi}{6} + i\sin\dfrac{\pi}{6}\right)$
$= 4\left(\cos\dfrac{\pi}{6} + i\sin\dfrac{\pi}{6}\right)$
$= 4\left(\dfrac{\sqrt{3}}{2} + i\dfrac{1}{2}\right) = 2\sqrt{3} + 2i$

Sol. 28. (c)
$(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$
$= (1 + \omega + \omega^2 + \omega^3)(1 + \omega^3.\,\omega)$
$[1 + (\omega^3)^2.\,\omega^2]$
$= 1.\,(1 + \omega)(1 + \omega^2)$
[$\because 1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$]
$= (1 + \omega + \omega^2 + \omega^3) = 1$

Sol. 29. (d)
We have,
$z = \dfrac{-2(1+2i)}{3+i}$
$= \dfrac{-2(2+2i)(3-i)}{(3+i)(3-i)}$
$= \dfrac{-2\left(3-i+6i-2i^2\right)}{9-i^2}$
$= \dfrac{-2}{10}(3+5i+2) = \dfrac{-10}{10}\left(i+1\right) = -i-1$
$\therefore$ Argument $(\theta) = -\pi + \theta = -\pi + \tan^{-1}(1/1)$
$= -\pi + \dfrac{\pi}{4} = \dfrac{-4\pi+\pi}{4} = \dfrac{-3\pi}{4}$

Sol. 30. (c)
Given, $i = \sqrt{-1}$
$\therefore \sqrt{i} = \sqrt{e^{i\frac{\pi}{2}}} = e^{i\frac{\pi}{4}}$

$= \cos\frac{\pi}{4} + i\sin\frac{\pi}{4}$

$= \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} = \frac{1+i}{\sqrt{2}}$

Sol. 31. (b)

$(\sin x + i\cos x)^3$

$= \left[\cos\left(\frac{\pi}{2} - x\right) + i\left(\sin\frac{\pi}{2} - x\right)\right]^3$

$= \cos 3\left(\frac{\pi}{2} - x\right) + i\sin 3\left(\frac{\pi}{2} - x\right)$

[By De Moivre's Theorem]

$= \cos\left(\frac{3\pi}{2} - 3x\right) + i\sin\left(\frac{3\pi}{2} - 3x\right)$

$= (-\sin 3x - i\cos 3x)$

Hence, the real part is $-\sin 3x$.

Sol. 32. (b)

$(x^3 - 1) = (x - 1)(x^2 + 1 + x) = (x - 1)(x^2 + x - \omega - \omega^2)$

$= (x - 1)(x^2 - \omega^2 + x - \omega)$

$= (x - 1)[(x + \omega)(x - \omega)(x - \omega)]$

$= (x - 1)(x - \omega)(x + \omega + 1)$

$= (x - 1)(x - \omega)(x - \omega^2)$

Sol. 33. (c)

$\left(\dfrac{\sin\frac{\pi}{6} + i\left(1 - \cos\frac{\pi}{6}\right)}{\sin\frac{\pi}{6} - i\left(1 - \cos\frac{\pi}{6}\right)}\right)^3$

$\Rightarrow \left(\dfrac{2\sin\frac{\pi}{12}\cos\frac{\pi}{12} + i\left(2\sin^2\frac{\pi}{12}\right)}{2\sin\frac{\pi}{12}\cos\frac{\pi}{12} - i\left(2\sin^2\frac{\pi}{12}\right)}\right)^3$

$\Rightarrow \left(\dfrac{\cos\frac{\pi}{12} + i\left(\sin\frac{\pi}{12}\right)}{\cos\frac{\pi}{12} - i\left(\sin\frac{\pi}{12}\right)}\right)^3 \Rightarrow \left(\dfrac{e^{i\frac{\pi}{12}}}{e^{-i\frac{\pi}{12}}}\right)^3$

$\Rightarrow \left(e^{i\frac{\pi}{6}}\right)^3 = e^{i\frac{\pi}{2}} = i$

Sol. 34. (d)

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

$|z_1| = \sqrt{x_1^2 + y_1^2}$

$|z_2| = \sqrt{x_2^2 + y_2^2}$

$|z_1| = |z_2|$

$\sqrt{x_1^2 + y_1^2} = \sqrt{x_2^2 + y_2^2}$

$x_1^2 + y_1^2 = x_2^2 + y_2^2$

Clearly, none of the options is correct

Sol. 35. (a)

Given equation of line is $i\bar{z} - iz = 5$.

or $i(\bar{z} - z) = 5$

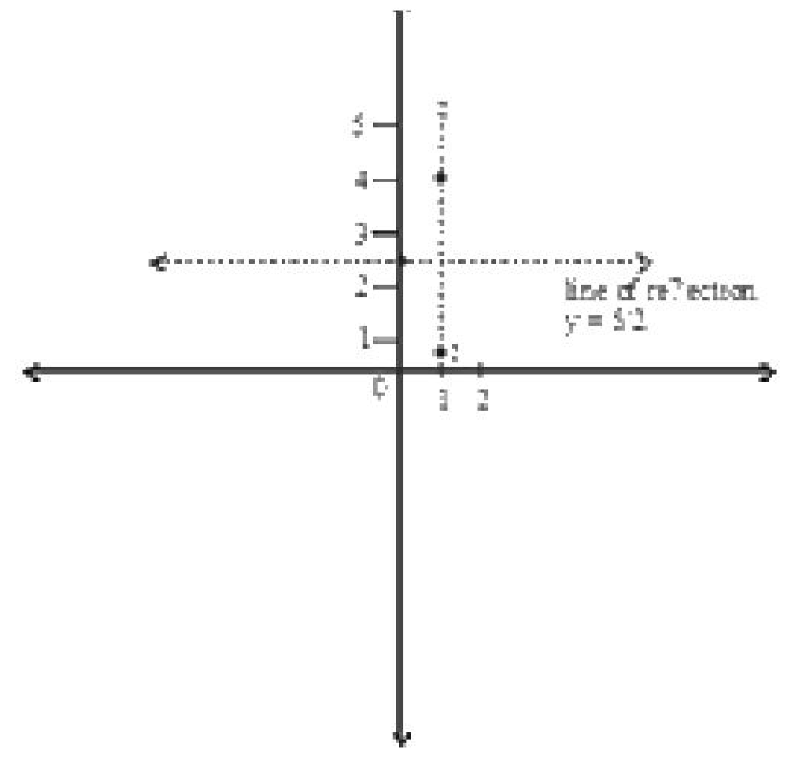
or $i(-2iy) = 5$

or $2y = 5$

or $y = 5/2$

if point $z_1 = 1 + i$

Then point $z_2 = 1 + 4i$



Sol. 36. (a)

Given,

$z\bar{z} + (3 - i)z + (3 + i)\bar{z} + 1 = 0$

Show (represent a circle)

$z = (x + iy)$ $\bar{z} = (x - iy)$

$(x + iy)(x - iy) + (3 - i)(x + iy) + (3 + i)(x - iy) + 1 = 0$

$x^2 + y^2 + 3x + 3iy - ix + i^2 y + 3x - ix + 3iy - i^2 y + 1 = 0$

$x^2 + y^2 + 6x + 2y + 1 = 0$

On comparing, $-3, -f = -1, c = 1$

$\therefore$ radius

$= \sqrt{g^2 + f^2 - c} = \sqrt{9 + 1 - 1} = \sqrt{9} = 3$

and coordinates of centre are $(-3, -1)$

Sol. 37. (c)

Let $z = x + iy$

Then from given relation $z^2 = i\bar{z}$

$(x+iy)^2 = i(x - iy)$

$x^2 - y^2 + 2ixy = y + ix$

$(x^2 - y^2 - y) + i(2xy - x) = 0$

$\Rightarrow x^2 - y^2 - y = 0$ and $2xy - x = 0$

from equation (ii)

$2xy - x = 0$

$x(2y - 1) = 0$

$x = 0, y = +\frac{1}{2}$

Putting $x = 0$ I equation (i)

$-y^2 - y = 0$

$-y(y+1) = 0$

$y = 0, y = -1$

Then the complex numbers are $0 + 0i$ and $0 - i$.

Putting $y = +\frac{1}{2}$ in equation (i)

$x^2 - \frac{1}{4} - \frac{1}{2} = 0$

$x^2 - \frac{3}{4} = 0$

$x^2 = \frac{3}{4}$

$x = \pm\frac{\sqrt{3}}{2}$

the complex number are

$\frac{\sqrt{3}}{2} + \frac{1}{2}i$ and $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$

Now non zero complex numbers are

$z_1 = -i$, $z_2 = \frac{\sqrt{3}}{2} + \frac{1}{2}i$

$z_3 = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$

$z_1 + z_2 + z_3 = 0$

Sol. 38. (c)

$z_1 . z_2 . z_3 = -i\left[\frac{\sqrt{3}}{2} + \frac{1}{2}i\right]\left[\frac{\sqrt{3}}{2} + \frac{1}{2}i\right]$

$= -i\left[-\frac{3}{2} + \frac{1}{4}i^2\right]$

$= -i\left[-\frac{3}{4} - \frac{1}{4}\right] = -i\left(\frac{-4}{4}\right)$

$\Rightarrow i$ (pure imaginary)

$z_1 z_2 + z_2 z_3 + z_3 . z_1 = -i\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) + \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$

$\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) + \left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)(-i)$

$= -\frac{\sqrt{3}}{2}i + \frac{1}{2} - \frac{3}{4} - \frac{1}{4} + \frac{\sqrt{3}}{2}i + \frac{1}{2}$

$= 1 - \frac{3}{4} - \frac{1}{4} = 0$

Sol. 39. (a)

Let $z = x + iy$

$\left|\frac{z - 4}{z - 8}\right| = 1$

$|z - 4| = |z - 8|$

$|z + iy - 4| = |x + iy - 8|$

$|(x-4) + iy|^2 = |(x - 8)^2 + iy|^2$

$(x - 4)^2 + y^2 = (x - 8)^2 + y^2$

$(x - 4)^2 = (x - 8)^2$

$x^2 + 16 - 8x = x^2 + 64 - 16x$

$-8x + 16x = 64 - 16$

$8x = 48$

$x = 6$

Again,

$\left|\frac{z}{z - 2}\right| = \frac{3}{2}$

$\frac{|z|}{|z - 2|} = \frac{3}{2}$

$2|z| = 3|z - 2|$

$4|z|^2 = 3|z - 2|^2$

$4|x + iy|^2 = 9|(x - 2) + iy|^2$

$4|x^2 + y^2| = 9|(x - 2)^2 + y^2|$

$4(x^2 + y^2) = 9[(x - 2)^2 + y^2]$

Putting $x = 6$

$4(36 + y^2) = 9(16 + y^2)$

$144 + 4y^2 = 144 + 9y^2$

$5y^2 = 0$

$y = 0$

Hence,

$z = 6 + 0i = 6$

$|z| = \sqrt{6^2 + 0^2} = 6$

Sol. 40. (d)

$\left|\frac{z - 6}{z + 6}\right| = \left|\frac{6 - 6}{6 + 6}\right| = 0$

Sol. 41. (c)

Since ω is a cube root of unity and $\omega \neq 1$. Then

$\omega = \frac{-1 + \sqrt{3}i}{2}, \omega^2 = \frac{-1}{2}, \frac{-\sqrt{3}i}{2}$

$\because$ Points P and Q are defined by ω and ω^2 respectively then

$P = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right), Q = \left(\frac{-1}{2}, \frac{-\sqrt{3}}{2}\right)$ and $O = (0, 0)$

Now, the gradient of line OP

$m_1 = \frac{y_2 - y_1}{x_2 - x_1}$

$m_1 = \dfrac{\frac{\sqrt{3}}{2} - 0}{-\frac{1}{2} - 0}$

$m_1 = - -\sqrt{3}$

Gradient of line OQ

$m_2 = \dfrac{-\frac{\sqrt{3}}{2} - 0}{-\frac{1}{2} - 0}$

$m_2 = \sqrt{3}$

Angle between lines OP and OQ

$\tan\theta = \dfrac{m_1 \sim m_2}{1 + m_1.m_2}$

$= \dfrac{\sqrt{3} - (-\sqrt{3})}{1 + (\sqrt{3})(-\sqrt{3})} = \dfrac{\sqrt{3}+\sqrt{3}}{1-3} = \dfrac{-2\sqrt{3}}{2}$

Or $\tan\theta = -\sqrt{3}$

$= -\tan 60 = \tan(180° - 60°)$

$= \tan 120°$

$\Rightarrow \theta = 120°$

Sol. 42. (d)

Since ω_1 and ω_2 are the roots of unity different from 1 then we get,

$\omega_1 = \dfrac{-1+\sqrt{3}i}{2}, \omega_2 = \dfrac{-1-\sqrt{3}i}{2}$

Now,

$(\omega_1 - \omega_2)^2 = \left(\dfrac{-1+\sqrt{3}i}{2} - \dfrac{-1-\sqrt{3}i}{2}\right)^2$

$= \left(\dfrac{-1+\sqrt{3}i+1+\sqrt{3}i}{2}\right) = \left(\dfrac{2\sqrt{3}i}{2}\right)^2$

$=3i^2 = -3$ $\{\because i^2 = -1\}$

Sol. 43. (a)

We have,

$z = x + iy = \left(\dfrac{1}{\sqrt{2}} - \dfrac{i}{\sqrt{2}}\right)^{-25}$

$= \left[\cos\left(\dfrac{\pi}{4}\right) + i\sin\left(-\dfrac{\pi}{4}\right)\right]^{-25}$

$= \cos\left(\dfrac{25\pi}{4}\right) + i\sin\left(\dfrac{25\pi}{4}\right)$

$= \cos\left(6\pi + \dfrac{\pi}{4}\right) + i\sin\left(6\pi + \dfrac{\pi}{4}\right)$

$= \cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}$

$\Rightarrow x + iy = \cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4} = \dfrac{1}{\sqrt{2}} + \dfrac{i}{\sqrt{2}}$

$\Rightarrow x = \dfrac{1}{\sqrt{2}}$ and $y = \dfrac{1}{\sqrt{2}}$

$\therefore \dfrac{z-\sqrt{2}}{z-i\sqrt{2}} = \dfrac{x+iy-\sqrt{2}}{x+iy-i\sqrt{2}}$

$= \dfrac{\frac{1}{2} + \frac{i}{\sqrt{2}} - \sqrt{2}}{\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} - i\sqrt{2}} = \dfrac{-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}}{\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}}$

$= -1$

$= -1 + 0.i$

$\therefore$ amplitude $= \tan^{-1}\left(\dfrac{0}{-1}\right) = \pi - \tan(0) = \pi$

Sol. 44. (d)

$\text{Re}\left(\dfrac{z-1}{z+1}\right) = 0$

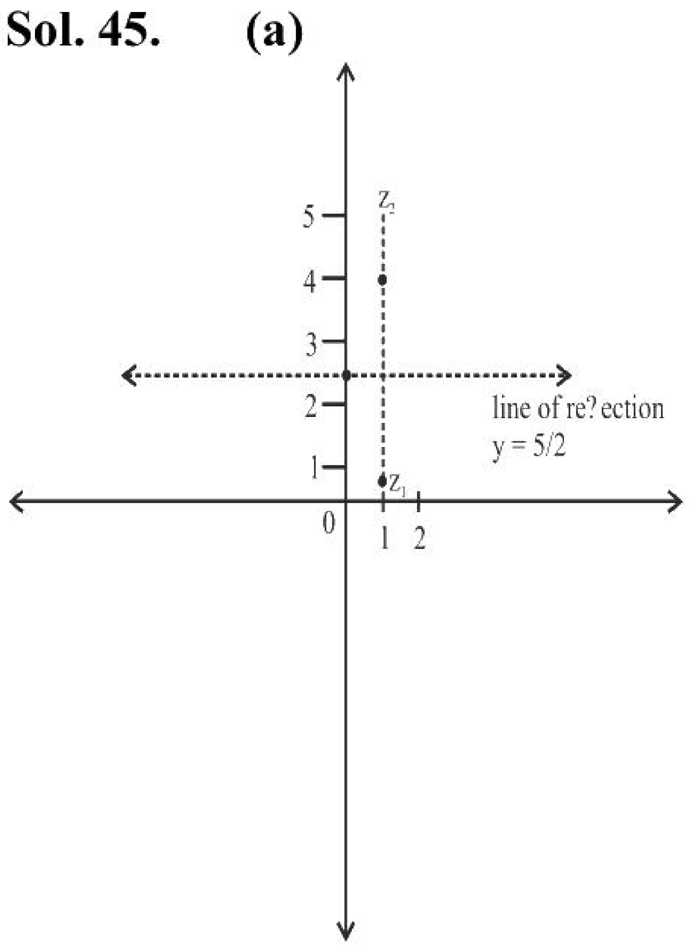
$\Rightarrow \text{Re}\left(\dfrac{(x-1)+iy}{(x+1)+iy} \times \dfrac{(x+1)-iy}{(x+1)-iy}\right) = 0$

$\Rightarrow \text{Re}\left(\dfrac{x^2-1+y^2+i(xy+y-xy+y)}{(x+1)^2+y^2}\right)$

$\Rightarrow x^2 + y^2 = 1$

$\Rightarrow |x| = 1$

Sol. 45. (a)



So imaginary part is zero.

Sol. 46. (c)

$z^2 + |z| = 0$

$(x+iy)^2 + \sqrt{x^2+y^2} = 0$

$x^2 + 2xyi - y^2 + \sqrt{x^2+y^2} = 0$

$x^2 - y^2 + \sqrt{x^2+y^2} + 2xyi = 0 + 0i$

$x^2 - y^2 + \sqrt{x^2+y^2} = 0$, $2xy = 0$

$2xy = 0 \Rightarrow x = 0$ or $y = 0$

if $y = 0$ than $x = 0$

if $x = 0$ than $y = 1,-1$

so there are 3 solutions (0,0),(0,1),(0,-1)

Sol. 47. (d)

$\omega^{100} + \omega^{200} + \omega^{300} = \omega + \omega^2 + 1 = 0$

Sol. 48. (b)

$\sqrt{\dfrac{1+\omega^2}{1+\omega}} = \sqrt{\dfrac{-\omega}{-\omega^2}} = \sqrt{\dfrac{1}{\omega}} = \sqrt{\omega^2} = \omega$

Sol. 49. (a)

We know that the sum

$i^x + i^{x+1} + i^{x+2} + i^{x+3} = 0$ {Here, $x = 2n$}

Sol. 50. (b)

$\left(\dfrac{-1+i\sqrt{3}}{2}\right)^n + \left(\dfrac{-1-i\sqrt{3}}{2}\right)^n = \omega^n + (\omega^2)^n$

$= \omega^n + \omega^{2n}$, where n is not a multiple of 3

or $n = 3k + r$, $r = 1$ or 2.

$\therefore \omega^2 + \omega^{2r} = \omega^{3k+r} + \omega^{2(3k+r)}$

$= \omega^{3k}.\omega^r + \omega^{6k}.\omega^r + \omega^{2r}$

Now, putting $r = 1$ or 2,

$\omega^n + \omega^{2n} = \omega + \omega^2 = -1.$

Sol. 51. (c)

$(1+\omega)(1+\omega^2)(1+\omega^3)(1+\omega+\omega^2) = 0$

$\{\because 1 + \omega + \omega^2 = 0\}$

Sol. 52. (a)

$z = \dfrac{1+2i}{1-(1-i)^2} = \dfrac{1+2i}{1-(1+i^2-2i)} = \dfrac{1+2i}{1+2i} = 1$

or $z = 1 + 0.i$

$\therefore$ Modulus (z) $= \sqrt{1^2+0^2} = 1.$

and argument (z) $= \tan^{-1}\left(\dfrac{0}{1}\right) = 0.$

Sol. 53. (c)

$|z + 4| \le 3$

$\Rightarrow -3 \le z + 4 \le 3$ $\{\because 0 \le |a| \le 3 \Rightarrow -3 \le a \le 3\}$

$\Rightarrow -7 \le z \le -1$

$\therefore$ The maximum value of

$|z + 1| = |-7 + 1| = 6$

Sol. 54. (c)

$z^2 = 2\bar{z}$

$\Rightarrow (x + iy)^2 = 2(x - iy)$

$\Rightarrow x^2 - y^2 + i2xy = 2x - i2y$

$\Rightarrow (x^2 - y^2 - 2x) + i2y(x + 1) = 0$

Comparing the coefficients on LHS and RHS,

$x^2 - y^2 - 2x = 0$

$\Rightarrow x^2 - 2x = y^2$ ……(i)

and $2y(x + 1) = 0$

$\Rightarrow y = 0$ or $x = -1$

If $y = 0$, then in equation (i),

$x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$

If $x = -1$, then in equation (i),

$-1(-3) = y^2 \Rightarrow y = \pm\sqrt{3}$

$\therefore$ The $z = 2 + i.0$

$-1 + \sqrt{3}i$ or $-1 - \sqrt{3}i$ or $0 + 0i$ or $2 + 0i$

Sol. 55. (b)

$\left(\dfrac{1+i}{1-i}\right)^n = 1$

or $\left[\dfrac{(1+i)(1+i)}{1-i^2}\right]^n = 1$

or $\left(\dfrac{1+i^2+2i}{2}\right)^n = 1$

or $i^n = 1 \Rightarrow n = 4.$

Sol. 56. (b)

$\left|z - \dfrac{4}{z}\right| \ge |z| - \left|\dfrac{4}{z}\right| \Rightarrow 2 \ge |z| - \dfrac{4}{|z|}$

$\Rightarrow |z|^2 - 2|z| - 4 \le 0$

$\Rightarrow |z| = \dfrac{2+2\sqrt{5}}{2} = 1 + \sqrt{5}$ (neglecting-ve value)

Sol. 57. (d)

All statements are correct

Sol. 58. (c)

Argument $(-1 - i)$

$\theta = \dfrac{\pi}{4}$

For third quadrant argument is

$\dfrac{\pi}{4} - \pi = -\dfrac{3\pi}{4}$

Sol. 59. (c)

Let $z = x + iy$

$\Rightarrow (x + iy)^2 + \alpha(x + iy) + \beta = 0$

$\Rightarrow x^2 - y^2 + 2xyi + \alpha x + i\alpha y + \beta = 0$

Equating real and imaginary parts separately, we get

$x^2 - y^2 + \alpha x + \beta = 0$, $(2x + \alpha)y = 0$

Now, $2x + \alpha = 0$ ($\because y \neq 0$)

Now, $1 - y^2 - 2 + \beta = 0$

$\Rightarrow \beta = 1 + y^2 > 1$, ($\because y \in R$, $y \neq 0$)

$\Rightarrow \beta \in (1, \infty)$

Sol. 60. (a)

$|1-2i|^x = 5^x$

$\Rightarrow 5^{x/2} = 5^x$
$\Rightarrow x = 0$
No any integral solution exists

Sol. 61. (c)

$$\left|\frac{\alpha-\beta}{|\alpha.\bar{\alpha}-\alpha\bar{\beta}|}\right| = \frac{|\alpha-\beta|}{|\alpha||\bar{\alpha}-\bar{\beta}|} = 1$$

Sol. 62. (a)

$i^{1000} + i^{1001} + i^{1002} + i^{1003} = 1 + i + i^2 + i^3$
$\Rightarrow 1 + i - 1 - i = 0$

Sol. 63. (b)

$z = \sqrt{3} + i$

$|z| = \sqrt{3+1} = 2$

$$\theta = \tan^{-1}\left|\frac{1}{\sqrt{3}}\right| = \frac{\pi}{6}$$

$$z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

Sol. 64. (c)

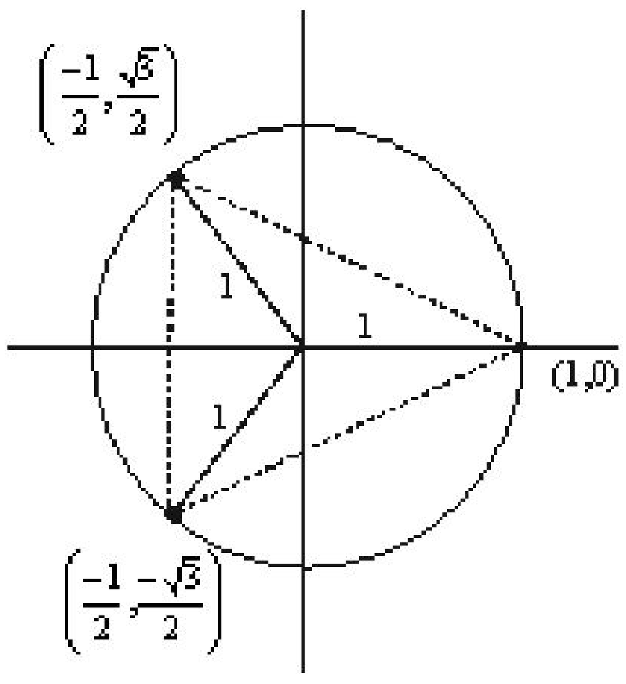
Sum $\Rightarrow i^2 + 2i^3 + 2i^4 + \ldots.. 2i^{10} + 2i^{11} + i^{12}$
$= 2i^{11} = 2i^3 = -2i$

Sol. 65. (b)

$(w)^{3n} + (w^2)^{3n}$
$(w^3)^n + (w^3)^{2n}$
$1 + 1 = 2$

Sol. 66. (c)

Δ is equilateral



Sol. 67.

ω and ω^2 are common roots of both equations.

Sol. 68. (d)

ω and ω^2 satisfies the equation

Sol. 69. (c)

$$\frac{1+2i}{1-(1-i)^2} = \frac{1+2i}{1+2i} = 1$$

modulus is 1

Sol. 70. (a)

argument of positive real number is 0

Sol. 71. (c)

$$\left[\frac{i(i+\sqrt{3})}{2i}\right]^{2019} + \left[\frac{i(i-\sqrt{3})}{2i}\right]^{2019}$$

$$\left[\frac{\omega}{i}\right]^{2019} + \left[\frac{\omega^2}{i}\right]^{2019} = \left[\frac{1}{i^3}\right] + \left[\frac{1}{i^3}\right]$$

$= 2i$

Sol. 72. (d)

Roots of equation $x^2 + x + 1 = 0$ are $\boldsymbol{\omega}$ and $\boldsymbol{\omega}^2$

$$\sum_{j=0}^{3}\left(\alpha^j + \beta^j\right) = \sum_{j=0}^{3}\left(\omega^j + \omega^{2j}\right)$$

$$[\omega^0 + \omega^{2(0)} + \omega^1 + \omega^2 + \omega^2 + \omega^4 + \omega^3 + \omega^6]$$

$= 1+1+(-1)+(-1)+1+1 = 2$

Sol. 73. (c)

$x = 1 + i$ then $x^2 = 2i$ and $x^4 = -4$
$x^6 + x^4 + x^2 + 1 = (x^4 + 1)(x^2 + 1)$
$= (2i + 1)(-4 + 1) = -6i - 3$

Sol. 74. (b)

$$\left|\frac{\cos\theta + i\sin\theta}{\cos\theta - i\sin\theta}\right| = \frac{|\cos\theta + i\sin\theta|}{|\cos\theta - i\sin\theta|}$$

= Let $z = \cos\theta + i\sin\theta$
then $\bar{z} = \cos\theta - i\sin\theta$
and we know that $|z| = |\bar{z}|$

so $$\frac{|\cos\theta + i\sin\theta|}{|\cos\theta - i\sin\theta|} = \frac{|z|}{|\bar{z}|} = 1$$

Sol. 75. (a)

$$\frac{1-i\sqrt{3}}{1+i\sqrt{3}} = \frac{\left(1-i\sqrt{3}\right)\left(1-i\sqrt{3}\right)}{\left(1+i\sqrt{3}\right)\left(1-i\sqrt{3}\right)}$$

$$\frac{\left(1-3-2i\sqrt{3}\right)}{4} = \frac{\left(-2-2i\sqrt{3}\right)}{4} = \frac{\left(-1-i\sqrt{3}\right)}{2}$$

$\tan\theta = 60°$
its lie in third quadrant so argument is $180°+60° = 240°$

Sol. 76. (a)

$zz^{-1} = 1$
second statement is wrong.

Sol. 77. (c)

$z + \bar{z} = 2\,\text{Re}(z)$

$z - \bar{z} = 2\,\text{Im}(z)$

Sol. 78. (b)

Modulus of any power of i is 1.

Sol. 79. (a)

$$\left(\frac{1-i}{1+i}\right)^{n^2} = \left(\frac{1-i}{1+i} \times \frac{1-i}{1-i}\right)^{n^2} = (-i)^{n^2} = 1$$

$n^2 = 4$
$n = 2$

Sol. 80. (b)

$z = 1 + i$

$$\frac{2}{z} = \frac{2}{1+i} \times \frac{1-i}{1-i} = 1 - i$$

$$\left|z + \frac{2}{z}\right| = |1 + i + 1 - i| = 2$$

Sol. 81. (c)

Roots of this equation are ω and ω^2
If one root is k than other is k^2

Sol. 82. (a)

$x^3 = 8$
$x = 2, 2w, 2w^2$
these roots are lie on a circle & 2 radius
2 unit on the ang and plane
These are lie on a circle so these are non- collinear

Sol. 83. (a)

$$\sum_{n=1}^{8n+7} = ?$$ where $i = \sqrt{-1}$

Here 8n is properly divisible by 4
So pertion left out will be $i + i^2 + i^3 + i^4 + i^5 + i^6 + i^7$
$= i - 1 - 1 + 1 + 1 - 1 - i$
$= -1$

Sol. 84. (c)

If $z = x + i\,y$ then $z\bar{z} + |z|^2 + 4\left(z + \bar{z}\right) - 48 = 0$

$(x + iy)(x - iy) + \left(\sqrt{x^2 + y^2}\right)^2 + 4(x + iy + x - iy) - 48 = 0$
$x^2 + y^2 + x^2 + y^2 + 4x - 48 = 0$
$2x^2 + 2y^2 + 4x - 48 = 0$
Or $x^2 + y^2 + 2x - 2y = 0$
Here h = 0 and coeff. x^2 and y^2 are
u a = b so it is a circle

Sol. 85. (a)

$2a + 2\sqrt{a^2 + b^2}$; a, b $\in$R
When you will do squaring of option a

$\left(\sqrt{a+ib} + \sqrt{a-ib}\right)^2$

$= a + ib + a - ib + 2\sqrt{(a+ib)(a-ib)}$

$= 2a + 2\sqrt{a^2 + b^2}$

So, ans is a.

Sol. 86. (b)

i^{-2n}; n$\in$Z

or $\frac{1}{i^{2n}} \Rightarrow \frac{1}{(i^2)^n} = \frac{1}{(-1)^n}$

if n is an even integer
then ans is 1
if n is an odd integer
then ans is -1

Sol. 87. (b)

$x^2 + x + 1 = 0$
$x^{199} + x^{200} + x^{201}$
$x^{199}(1 + x + x^2) = x^{199} \times 0 = 0$

Sol. 88. (b)

$$\frac{1}{1+i} = \frac{1-i}{2} = \frac{1}{2} - \frac{i}{2} =$$

$\tan\theta = \left|\frac{-1/2}{1/2}\right| = 1$

$\theta = \pi/4$

Avg.(z) = $\frac{-\pi}{4}$

Sol. 89. (c)

$$z = \left(\frac{-1}{2} + \frac{\sqrt{3}i}{2}\right)^{200}$$

$z = \omega^{200}$
$|z| = |\omega^{200}| = |\omega|^{200} = 1$

Sol. 90. (a)

$$\left|\frac{1+i\sin\theta}{1-i\sin\theta}\right| = \frac{|1+i\sin\theta|}{|1-i\sin\theta|} = \frac{\sqrt{1+\sin^2\theta}}{\sqrt{1+\sin^2\theta}} = 1$$

Sol. 91. (c)

$$\frac{1+i\sin\theta}{1-i\sin\theta} \times \frac{1+i\sin\theta}{1+i\sin\theta} \Rightarrow \frac{1+2i\sin\theta - \sin^2\theta}{1+\sin^2\theta}$$

For purely real Im(z) = 0

$$\Rightarrow \frac{2i\sin\theta}{1+\sin^2\theta} = 0$$

$\Rightarrow \sin\theta = 0 \Rightarrow \theta = n\pi$

Sol. 92. (b)

$$\Rightarrow \frac{1+2i\sin\theta - \sin^2\theta}{1+\sin^2\theta}$$

For purely imaginary Re(z) = 0

$$\Rightarrow \frac{1-\sin^2\theta}{1+\sin^2\theta} = 0 \Rightarrow 1 - \sin^2\theta = 0$$

$$\Rightarrow \cos\theta = 0 \Rightarrow \theta = \frac{(2n+1)}{2}\pi$$

Sol. 93. (c)

$$\frac{z-1}{z+1} \Rightarrow \frac{x+iy-1}{x+iy+1}$$

$$\Rightarrow \frac{(x-1)+iy}{(x+1)+iy} \times \frac{(x+1)-iy}{(x+1)-iy}$$

For purely imaginary Re(z) = 0

$$\Rightarrow \frac{(x-1)(x+1)+y^2}{(x+1)^2+y^2} = 0$$

$\Rightarrow x^2+y^2-1=0$ or $\Rightarrow x^2+y^2=1$

$\Rightarrow |z|=1$

Sol. 94. (b)

$z=\frac{1+i\sqrt{3}}{1-i\sqrt{3}}$

Argument of $1+i\sqrt{3}$ is $\frac{\pi}{3}$

Argument of $1-i\sqrt{3}$ is $-\frac{\pi}{3}$

$\arg\left(\frac{z_1}{z_2}\right)=\arg(z_1)-\arg(z_2)$

$\arg\left(\frac{z_1}{z_2}\right)=\frac{\pi}{3}-\left(-\frac{\pi}{3}\right)=\frac{2\pi}{3}$

Sol. 95. (a)

$\left|\frac{1-\omega}{\omega+\omega^2}\right|=\left|\frac{1-\omega}{-1}\right|=|1-\omega|$

$=\left|1-\frac{-1+\sqrt{3}i}{2}\right|=\left|\frac{3+\sqrt{3}i}{2}\right|=\frac{\sqrt{9+3}}{2}=\sqrt{3}$

Sol. 96. (d)

$\left|\frac{\alpha^{100}+\beta^{100}}{\alpha^{100}-\beta^{100}}\right|=\left|\frac{(-\omega)^{100}+(-\omega^2)^{100}}{(-\omega)^{100}-(-\omega^2)^{100}}\right|$

$\Rightarrow\left|\frac{\omega+\omega^2}{\omega-\omega^2}\right|=\left|\frac{-1}{-1-2\omega^2}\right|$

$\Rightarrow\left|\frac{1}{-1-2\left(\frac{-1-\sqrt{3}i}{2}\right)}\right|=\left|\frac{1}{\sqrt{3}i}\right|=\frac{1}{\sqrt{3}}$

Sol. 97. (d)

$z\bar{z}=|z+\bar{z}|$

$x^2+y^2=|x+iy+x-iy|$

$x^2+y^2=\pm 2x$

$x^2+y^2\pm 2x=0$

Equation of circles.

Sol. 98. (c)

$\sqrt{12+5i}+\sqrt{12-5i}$

$\Rightarrow\sqrt{\left(\sqrt{12+5i}+\sqrt{12-5i}\right)^2}$

$\Rightarrow\sqrt{\left(12+5i+12-5i+2\sqrt{12+5i}\sqrt{12-5i}\right)}$

$\Rightarrow\sqrt{\left(24+2\sqrt{144+25}\right)}=\sqrt{(24+26)}=5\sqrt{2}$

Sol. 99. (b)

Let $z=-i=0-i=r(\cos\theta+i\sin\theta)$ …(i)

Modulus of $-i=1$

Argument of $-i=-90^0$

Complex number in polar form is

$\left\{\cos\left(\frac{\pi}{2}\right)-i\sin\left(\frac{\pi}{2}\right)\right\}$

$\sqrt{z}=\sqrt{-i}=\pm(1)^{1/2}\left(\cos\frac{\pi}{2}-i\sin\frac{\pi}{2}\right)^{1/2}$

$=\frac{1-i}{\sqrt{2}}$

Sol. 100. (b)

$z^3+2z^2+2z+1=0$

$(z+1)(z^2+z+1)=0$

$z=-1$ or $z=\omega, \omega^2$

Sol. 101. (d)

By checking options

ω is satisfying the equation

$\omega^{1985}+\omega^{100}+1=\omega^2+\omega+1=0$

Sol. 102. (c)

Check all three roots of equation – 1

-1 not satisfy equation 2

ω and ω^2 satisfy equation 2

So there will be 2 common roots.

Sol. 103. (b)

$(z-100)^3+1000=0$

$(z-100)^3=-1000$

$z-100=-10$ or -10ω or $-10\omega^2$

$z=90$, or $100-10\omega$, or $100-10\omega^2$

Sol. 104. (c)

$(1+i)^4+(1-i)^4$

$[(1+i)^2]^2+[(1-i)^2]^2$

$(2i)^2+(-2i)^2=-2-2=-4$

Sol. 105. (b)

$iz^3+z^2-z+i=0$

$iz^3+z^2+i^2z+i=0$

$z^2(iz+1)+i(iz+1)=0$

$(iz+1)(z^2+i)=0$

if $z^2=-i$

$|z^2|=|-i|=1$

$|z|=1$

So $[|z|+1]^2=2^2=4$

Sol. 106. (a)

Cube root of unity are $1, \omega, \omega^2$

$x=1, y=\omega, z=\omega^2$

$xy+yz+zx=1+\omega+\omega^2=0$

Sol. 107. (c)

$\left|\frac{z_1+z_2}{z_1-z_2}\right|=1$

let $z_1=x_1+iy_1$ and $z_2=x_2+iy_2$

$\frac{|(x_1+iy_1)+(x_2+iy_2)|}{|(x_1+iy_1)-(x_2+iy_2)|}=1$

$\frac{|(x_1+x_2)+i(y_1+y_2)|}{|(x_1-x_2)+i(y_1-y_2)|}=1$

$|(x_1+x_2)+i(y_1+y_2)|=|(x_1-x_2)+i(y_1-y_2)|$

$\sqrt{(x_1+x_2)^2+(y_1+y_2)^2}=\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

$x_1x_2+y_1y_2=0$

now

$\text{Re}\left(\frac{z_1}{z_2}\right)+1$

$\text{Re}\left(\frac{x_1+iy_1}{x_2+iy_2}\right)+1\Rightarrow\text{Re}\left(\frac{x_1+iy_1}{x_2+iy_2}\times\frac{x_2-iy_2}{x_2-iy_2}\right)+1$

$\text{Re}\left(\frac{(x_1x_2+y_1y_2)+i(x_2y_1-x_1y_2)}{x_2^2+y_2^2}\right)+1=0+1=1$

$\left(\frac{x_1x_2+y_1y_2}{x_2^2+y_2^2}\right)+1=0+1=1$

Sol. 108. (a)

$Z_1^2+Z_2^2+Z_1Z_2=0.$

$\left(\frac{Z_1}{Z_2}\right)^2+\left(\frac{Z_1}{Z_2}\right)+1=0$

let $\left(\frac{Z_1}{Z_2}\right)=x$

$x^2+x+1=0$

roots of above equation are ω and ω^2

so $\left(\frac{Z_1}{Z_2}\right)=\omega$ and $\left(\frac{Z_1}{Z_2}\right)=\omega^2$

$\left|\frac{Z_1}{Z_2}\right|=|\omega|=|\omega^2|=1$

Sol. 109. (b)

by above solution

$\frac{1}{2}+\text{Re}\left(\frac{Z_1}{Z_2}\right)=\frac{1}{2}+\text{Re}(\omega)$

$=\frac{1}{2}+\text{Re}\left(\frac{-1+i\sqrt{3}}{2}\right)=\frac{1}{2}-\frac{1}{2}=0$

Sol. 110. (d)

$(1+\omega-\omega^2)^{100}+(1-\omega+\omega^2)^{100}$

$(-\omega^2-\omega^2)^{100}+(-\omega-\omega)^{100}$

$(-2\omega^2)^{100}+(-2\omega)^{100}$

$(2)^{100}(\omega^{200}+\omega^{100})=(2)^{100}(\omega^2+\omega^1)=-2^{100}$

Sol. 111. (b)

$z=\frac{1}{3}\begin{vmatrix} i & 2i & 1 \\ 2i & 3i & 2 \\ 3 & 1 & 3 \end{vmatrix}=x+iy;$

$z=1+i=x+iy$

$|z|=|1+i|=\sqrt{1+1}=\sqrt{2}$

Sol. 112.

$\sum_{n=1}^{20}(i^{n-1}+i^n+i^{n+1})$

$\sum_{n=1}^{20}(i^{n-1})+\sum_{n=1}^{20}(i^n)+\sum_{n=1}^{20}(i^{n+1})$

$=0+0+0=0$

Sol. 113. (a)

$\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^3=\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\times\frac{i}{i}\right)^3$

$\left(\frac{\sqrt{3}i-1}{\sqrt{3}i+1}\right)^3=\left(\frac{2\omega}{-2\omega^2}\right)^3=\left(\frac{1}{-\omega}\right)^3=-1$

Sol. 114. (c)

$x^2-x+1=0$ or $x^2+1=x$

$x^2+1=x$ or $\left(x+\frac{1}{x}\right)=1$

$(\alpha+\beta)^2-(\alpha-\beta)^2=4\alpha\beta$

$\left(x+\frac{1}{x}\right)^2-\left(x-\frac{1}{x}\right)^2=4$

$1-\left(x-\frac{1}{x}\right)^2=4$ or $x-\frac{1}{x}=\sqrt{3}i$

$(\sqrt{3}i)^2+(\sqrt{3}i)^4+(\sqrt{3}i)^8=-3+9+81=87$

Sol. 115. (a)

$amp(\bar{z})=-amp(z)$

So $amp(z)+amp(\bar{z})=0$

Sol. 116. (c)

We know that $\alpha=\frac{-1+\sqrt{-3}}{2}=\omega$

$(1+\omega^{19}-\omega^{35})^{100}-(1-3\omega^{25}+\omega^{38})^{50}$

$(1 + \omega - \omega^2)^{100} - (1 - 3\omega + \omega^2)^{50}$
$(-\omega^2 - \omega^2)^{100} - (-\omega - 3\omega)^{50}$
$(-2\omega^2)^{100} - (-4\omega)^{50}$
$2^{100}\omega^{200} - 4^{50}\omega^{50} = 2^{100}\omega^2 - 4^{50}\omega^2 = 0$

Sol. 117. (b)

$$\frac{1+i}{1-i}\times\frac{1+i}{1+i}=\frac{(1+i)^2}{2}=i$$

$$\left(\frac{1-i}{1+i}\right)^{2m}\left(\frac{1+i}{1-i}\right)^{2n}=1$$

$$\left(\frac{1}{i}\right)^{2m}(i)^{2n}=1$$

$$(-i)^{2m}(i)^{2n}=1$$

$$(-1)^m(-1)^n=1$$

m and n should be both odd or both even.
And smallest difference between two odd numbers or two even numbers is 2.

Sol. 118. **(a)**

$x^2 + y^2 + z^2$
$= (a + b)^2 + (a\omega + b\omega^2)^2 + (a\omega^2 + b\omega)^2$
$= a^2 + 2ab + b^2 + a^2\omega^2 + 2ab\omega^3 + b^2\omega^4 + a^2\omega^4 + 2ab\omega^3 + b^2\omega^2$
$= a^2(1 + \omega + \omega^2) + 6ab + b^2(1 + \omega + \omega^2)$
$= 6ab$

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